# Isomerism

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# Degree of Unsaturation (DU)

**INTRODUCTION**

The main objective of an organic chemist is the determination of the structure of a new organic compound which has been obtained in pure state either from a natural source or synthesised in the laboratory.

In order to establish the correct structure of an organic compound, it is necessary to detect skeleton of compound, elements and functional groups present in the organic compound.

$$C_nH_{2n+2} \xrightarrow{-2H} C_nH_{2n} \xrightarrow{-2H} C_nH_{2n-2}$$

| saturated open chain alkane (DU = 0) | unsaturated (DU = 1) | unsaturated (DU = 2) |
|---|---|---|

Deficiency of hydrogen generates unsaturation in the compound to compute this unsaturation is called DU.

How many bond cleavage give open chain saturated compound is DU of that compound.

Open chain saturated compound having DU = 0.

In compound unsaturation is observed in the form of $\pi$-bonds & rings.

**Tips and Trick**

DU = 1 → Ring' or Double bond'

DU = 2 → '=' bond + '=' bond

→ '≡' bond

→ '=' bond + ring

→ Ring + Ring

**Case-I :** If molecular formula is given.

$$DU = (C+1) - \left[\frac{H + X - N}{2}\right]$$

(Where C = carbon, H = hydrogen, N = nitrogen)

**i)** $C_4H_8$

$(4+1) - \frac{8}{2} = 5 - 4 \Rightarrow DU = 1$

**ii)** $C_4H_6$

$(4+1) - \frac{6}{2} = 5 - 3 = 2$

**iii)** $C_6H_6$

$(6+1) - \frac{6}{2} = 7 - 3 = 4$

**iv)** $C_6H_6O$

$(6+1) - \frac{6}{2} = 7 - 3 = 4$

**v)** $C_8H_8$

$$(8+1)-\frac{8}{2} = 9 - 4 = 5$$

**vi)** $C_8H_8O$

$$(8+1)-\frac{8}{2} = 9 - 4 = 5$$

**vii)** $C_8H_4BrFClI$

$$(8+1)-\left(\frac{4+4}{2}\right) = 9 - 4 = 5$$

**viii)** $C_5H_9N$

$$(5+1)-\left(\frac{9-1}{2}\right) = 6 - 4 = 2$$

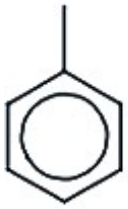
**ix)** $C_3H_9N$

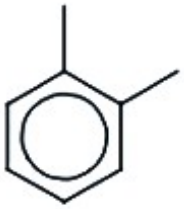
$$(3+1)-\left(\frac{9-1}{2}\right) = 4 - 4 = 0$$

**x)** $C_8H_3BrFNO$

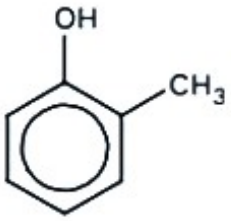
$$(8+1)-\left(\frac{3+2-1}{2}\right) = 9 - 2 = 7$$

**Case-II :** If molecular formula is given.
DU = Number of $\pi$-bond + number of rings

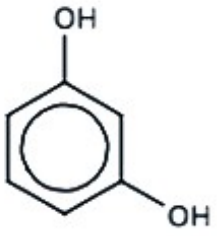
**i)**  Toulene (4)

**ii)**  o-xylene (4)

**iii)**  p-xylene (4)

**iv)** 

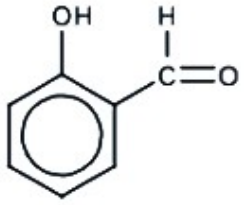
o-cresol (4)

**v)** 

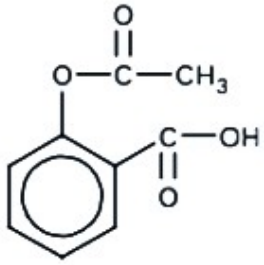
Resorcinol (4)

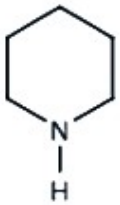
**vi)** 

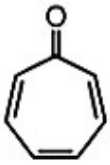
p-toluidine (4)

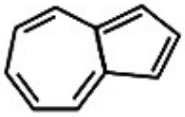
vii)  o-salicyaldehyde (5)

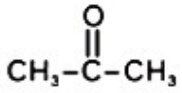
viii)  Anisol (4)

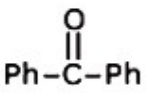
ix)  Aspirin (6)

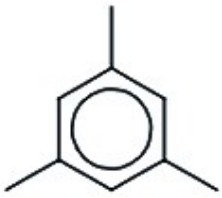
x)  Piperidine (1)

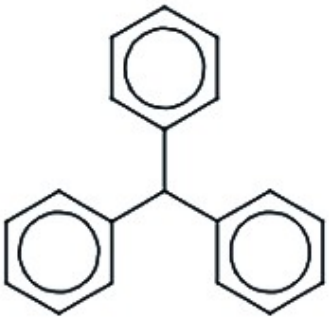
xi)  Tropolone (5)

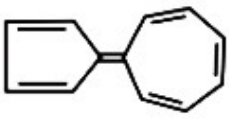
xii)  Azulene (7)

xiii)  Acetone (1)

xiv)  Benzo phenone (9)

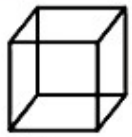
xv)  Mesitylene (4)

xvi)  Triphenyl methane (12)

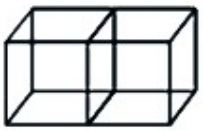
xvii)  (8)

xviii)  (7)

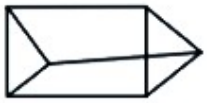
**Case-III :** If molecular structure is given in 3-D
DU = number of faces – number of box

i) 
Cubane

$C_8H_8$
DU = 6 – 1 = 5

ii) 

$C_{12}H_8$
DU = 11 – 2 = 9

iii) 

$C_6H_6$
DU = 5 – 1 = 4

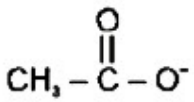
iv) 

2

v) 

3

**Note :** in case of salt DU is fractional.
**Example :**



$$DU = (C+1) - \left(\frac{H+X-N}{2}\right)$$

$$= (2+1) - \left(\frac{3}{2}\right) = 1.5$$

# Introduction of Isomerism and Structural Isomers

Iso → same

mers → unit

**Example :** $C_2H_6O$ $\underbrace{CH_3CH_2OH \ \& \ CH_3OCH_3}_{\text{Isomers}}$

**Definitions**

- Two or more different compounds having same molecular formula are known as isomers.

**CHARACTERISTIC OF ISOMERS**

- They have same molecular formula.
- They have same molecular weight.
- They have same emperical formula.
- Emperical formula : Formula in simplest ratio or in condense form.

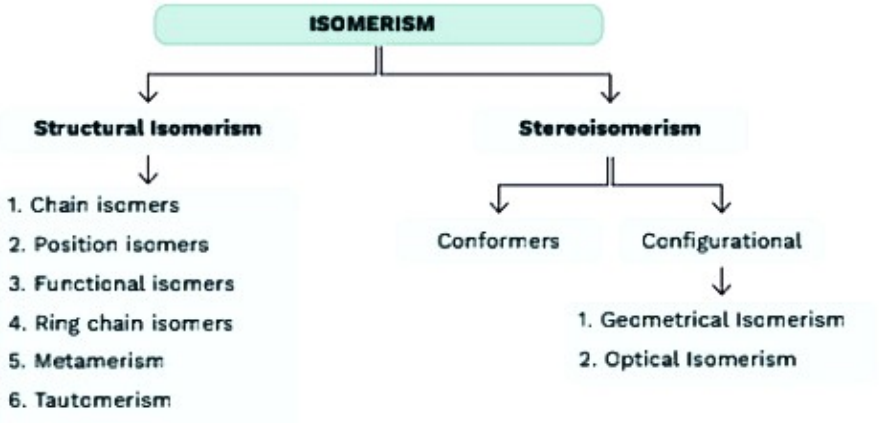
  Eg. MF ⇒ EF

  $C_6H_6$ ⇒ CH

  $C_6H_{12}O_6$ ⇒ $CH_2O$

- They have same emperical weight.
- They have same vapour density.

  2 × vapour density = Molecular weight
- They have same DU.
- Atleast one physical or chemical property must be different.

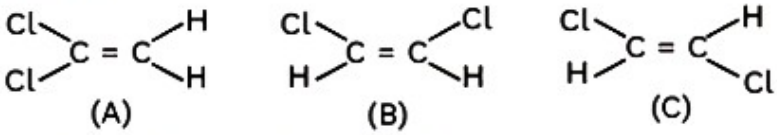



**STRUCTURAL ISOMERS**

Two or more different compound having same molecular formula but have difference in connectivity of atoms or groups.

## STEREOISOMERS

**Example :**



(A) (B) (C)

(A, B) ⇒ Structural, (A, C) ⇒ Structural,
(B, C) ⇒ Stereoisomers

Boiling point ∝ Dipole moment

Dipole moment of B ⇒ $\mu \neq 0$

Dipole moment of C ⇒ $\mu = 0$

Dipole moment and boiling point are physical property. So here B & C have different physical properties.

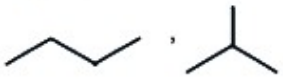
**Definitions**

- Two or more different compound having same molecular formula as well as structural formula but have difference in orientation of atoms & groups in space or 3-D.

## STRUCTURAL ISOMERS

**1. Chain Isomers**

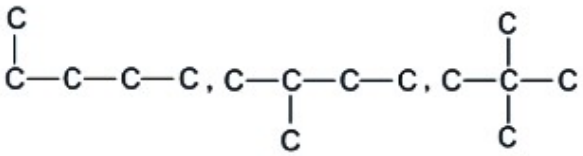
**Example :**

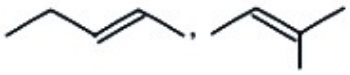
i) 

ii) 

iii) 

iv) $CH_3 - CH_2 - CH_2 - CN$, $CH_3 - \underset{\displaystyle CN}{\underset{|}{CH}} - CH_3$

v) 

vi) 

**Definitions**

- Two or more different compound having same molecular formula but different in parent carbon chain or side chain.

**Points to remember**

- Chain isomers must have same functional group.

## 2. Positional Isomers

- For positional isomers, parent carbon chain must be same.
- For positional isomers, functional group remain same.

**Definitions**

- Two or more different compound having same molecular formula but difference in position of substituent, multiple bond and functional group are known as positional isomers.

**Examples (Solved) :**

i) 

Positional isomers ✓

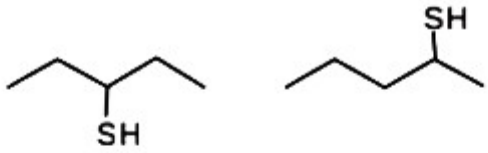
ii) 

Chain isomers ✓

They are chain isomers because PCC is not same

iii) 

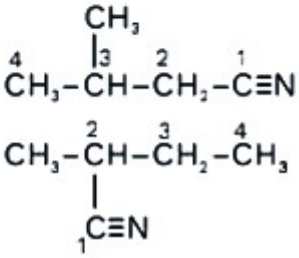
Positional isomers ✓

iv) 


Positional isomers ✓

v) 

Positional isomers ✓

vi) 


Positional isomers ✓



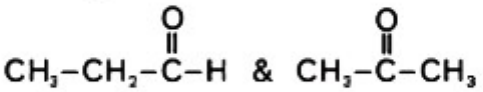

## 3. Functional Isomers

**Examples (Solved) :**

- Alcohol – ether

  $CH_3 - CH_2 - OH$ & $CH_3 - O - CH_3$

- Aldehyde – ketone




**Definitions**

- Two or more different compound having same molecular formula but different functional groups.

- Alkyne – diene

  $CH \equiv C - CH_3$ $CH_2 = C = CH_2$

- 1°, 2°, 3° amine are different functional group.

  $CH_3 - CH_2 - CH_2 - CH_2 - NH_2$, $CH_3 - CH_2 - NH - CH_2 - CH_3$, $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{N}} - CH_2 - CH_3$

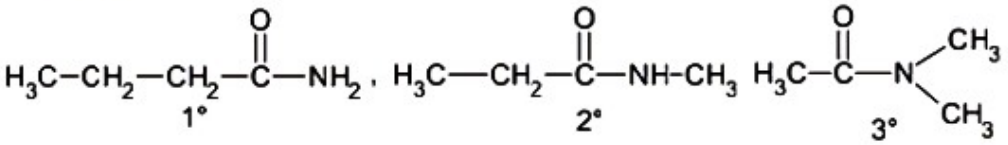
- Nitrile & isonitrile

  $CH_3 - C \equiv N$ or $CH_3 - N \equiv C$

**Points to remember**

- 1°, 2°, 3° amide are different functional group

  **Example :**



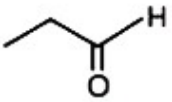
# Structural Isomerism

**RING CHAIN ISOMERS**

**Examples :**

1.  alkane  alkene

   Ring chain isomers or functional isomers

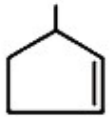
2.  Ether  aldehyde

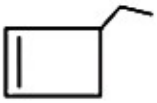
   Ring chain isomers or functional isomers

**Definitions**

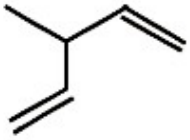
- Two or more different compound having same molecular formula but one is chain & another one is close chain with same DU.

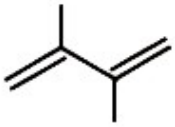
3. Find relations in (a & b), (a & d), (a & f), (b & c), (b & e), (b & g), (c & e), (c & g), (d & f) and (e & g) :

i) 

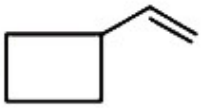
ii) 

iii) 

iv) 

v) 

vi) 

vii) 

**Ans.**

(a) & (b) - chain isomer
(a) & (d) - ring chain
(a) & (f) - ring chain & functional isomer
(b) & (c) -ring chain & functional isomer
(b) & (e) - ring chain
(b) & (g) - ring chain
(c) & (e) - functional isomer
(c) & (g) - functional isomer
(d) & (f) - functional isomer
(e) & (g) - chain isomer

## METAMERISM

### Monovalent functional groups

1. $-OH$
2. $-NH_2$
3. $-\overset{\overset{O}{\|}}{C}-NH_2$
4. $-\overset{\overset{O}{\|}}{C}-OH$
5. $-\overset{\overset{O}{\|}}{C}-H$
6. $-\overset{\overset{O}{\|}}{C}-X$
7. $-\overset{|}{N}-$

### Polyvalent functional groups

1. $-O-$
2. $-S-$
3. $-\overset{\overset{O}{\|}}{C}-$
4. $-\overset{\overset{O}{\|}}{C}-O-$
5. $-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-O-$
6. $-NH-$
7. $-\overset{\overset{O}{\|}}{C}-\overset{|}{N}-$
8. $-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-$

**Definitions**

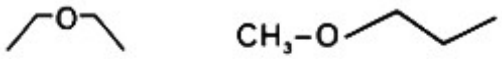
- If alkyl or aryl group changes around polyvalent functional group then these isomers are known as metamers.

**Points to remember**

- Those group which have more than one free valency are known as polyvalent functional groups.

**Examples (Solved) :**

Find relation between given isomers :

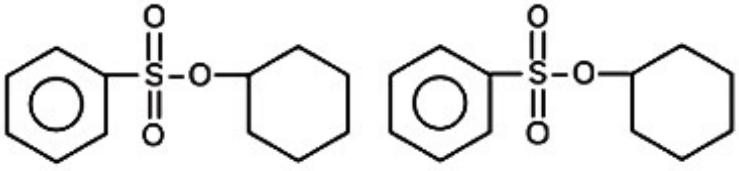
i)  Metamers

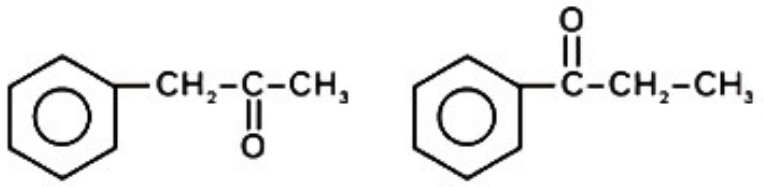
Metamers

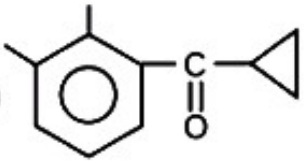
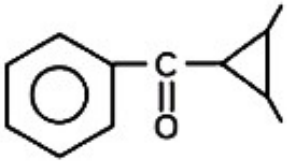
ii)  Identical

Identical

iii)  Positional isomers

iv)  Metamers

v)  Metamers

vi) (a)  (b)  Metamers

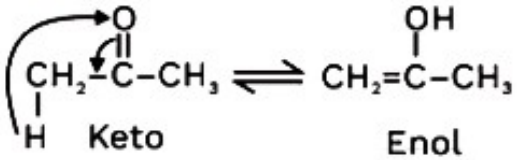
**Points to remember**

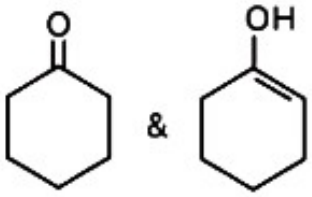
- Functional group remain same for metamerism.

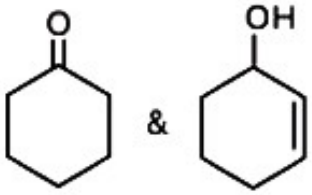
**TAUTOMERISM**

Protonotropy / cationotropy / desmotropism / keto enol tautomerism

**Examples :**

i) 


ii)  Tautomers & functional isomers

iii)  Functional isomer only

**Definitions**

- Those functional isomers which are present in dynamic equilibrium are called tautomers. This process is known as Tautomerism.

**Introduction**

- As in tautomerism, proton migrates, therefore it is also known as protonotropy.
- In tautomerism bonds also migrate, therefore it is also known as desmotropism.
- For tautomerism atleast one $\alpha$-hydrogen is required.

**Tips and Trick**

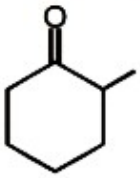
**Priority order :**

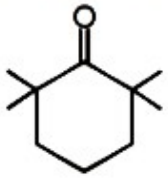
R > C > T > F > M > C > P

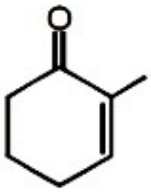
Radio city top FM near Carrer point

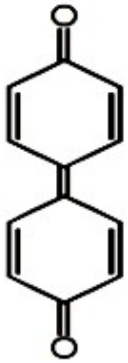
**Examples (Solved) :**

Among the following compounds how many will show tautomerism ?

i)  Yes

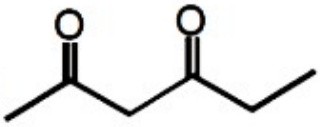
ii)  No

iii)  Yes

iv)  No

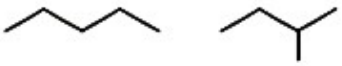
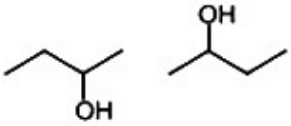
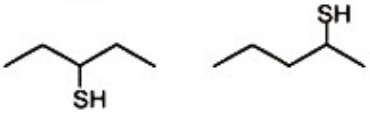
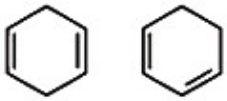
v) 2-pentanone Yes

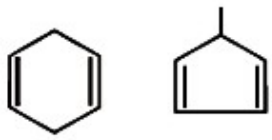
vi) Acetone Yes

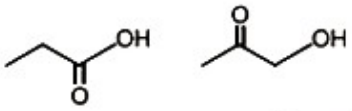
vii)  Yes

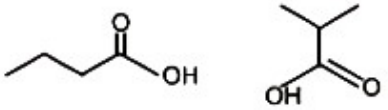
**Examples (Solved) :**

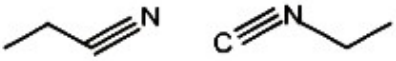
Define relation between the following:

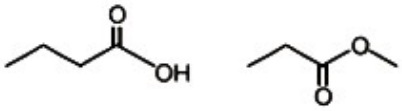
1.  Chain isomers
2.  Different compound
3.  Different compound
4.  Functional isomer
5.  Identical
6.  Identical
7.  Positional isomer
8.  Positional isomer

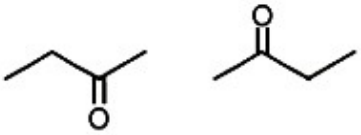
9.  Chain isomers

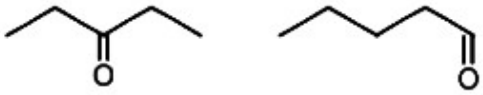
10.  Functional isomer

11.  Chain isomers

12.  Functional isomer

13.  Functional isomer

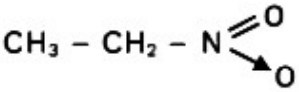
14.  Identical

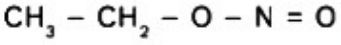
15.  Functional isomer

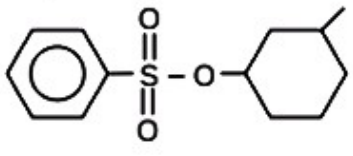
16.  Metamers & positional isomer

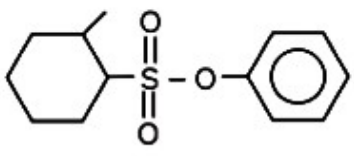
17.  Functional isomer

18.  Chain isomer

19. 

 Functional isomer

20. 

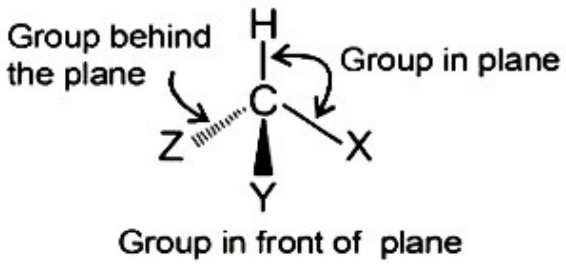
 Metamers

# Projection Formulas in Organic Chemistry

**Introduction**

The configuration of organic molecules can be visualised by three-dimensional (3D) structures, which may be depicted by any of the following representations :

## WEDGE-DASH FORMULA



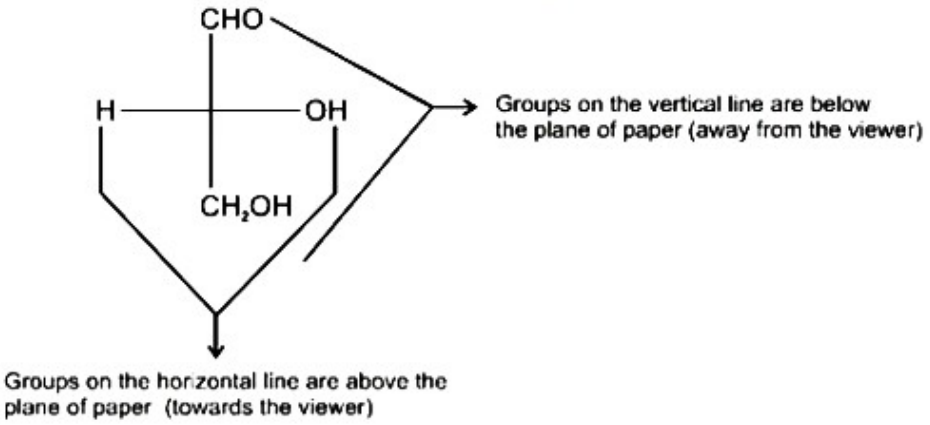

The thick solid (solid wedge) line indicates the bonds lying above the plane of paper (projecting towards the viewer).

Thin lines indicate the bonds lying within the plane of paper.

The dotted lines indicate the bonds lying below the plane of paper. Such a representation is called Wedge-Dash formula.

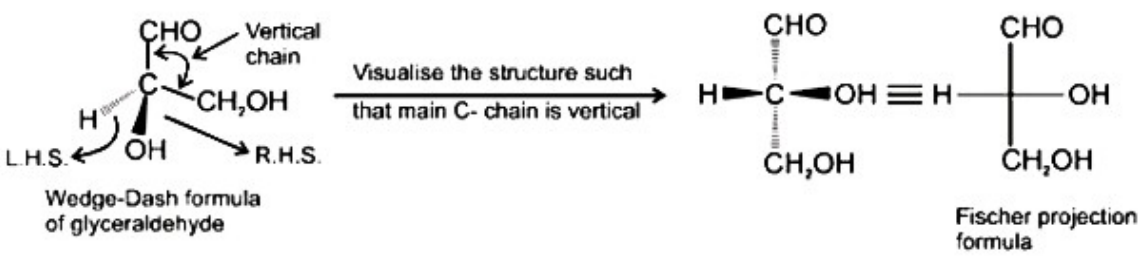
## FISCHER PROJECTION FORMULA

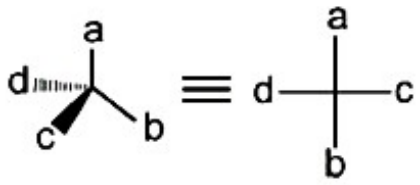
The horizontal lines represent the bonds directed towards the viewer and the vertical lines away from the viewer. For example, (–CHO) group in glyceraldehyde, (–COOH) group in lactic acid etc. The C atoms are then numbered according to the IUPAC nomenclature.



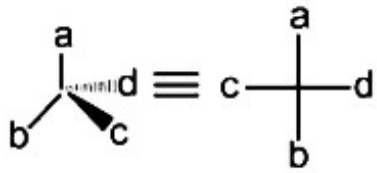

## INTERCONVERSION OF WEDGE-DASH FORMULA TO FISCHER FORMULA

**For Example:**

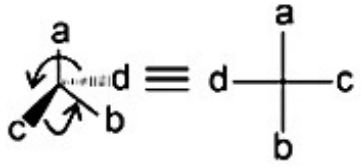


i) 

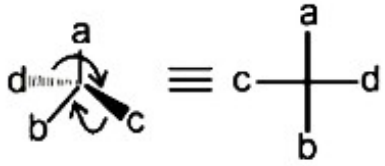
(d on L.H.S.)
and
(c on R.H.S.)

ii) 

(c on L.H.S.)
and
(d on R.H.S.)

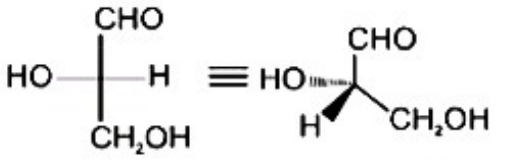
iii) 

(d on L.H.S.)
and
(c on R.H.S.)

iv) 

(d on R.H.S.)
and
(c on L.H.S.)

**INTERCONVERSION OF FISCHER FORMULA TO WEDGE-DASH FORMULA**



Fischer formula  Wedge-Dash formula

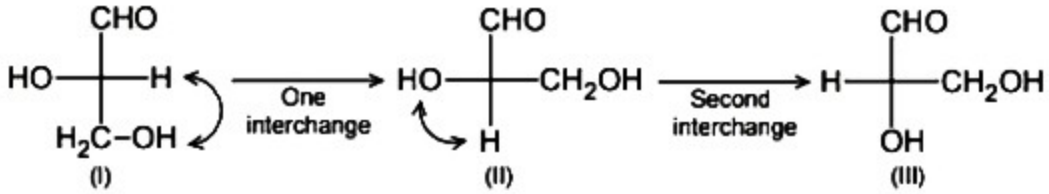
**THE FOLLOWING RULES MUST BE OBSERVED**

1. A fischer projection should not taken out of the plane of paper and turned over. It would lead to the Fischer projection of the enantiomer, since it changes the configuration at the chiral centre.



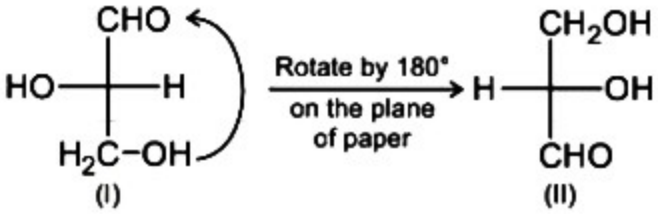
Enantiomer

2. Even number of interchanges clockwise or anti-clockwise leads to the same stereoisomer, while odd number of interchanges results in an enantiomer.




(1) and (3) are equivalent Fischer projections.

3. Rotation of Fischer projection by 180º on the plane of paper results in the equivalent Fischer projection.




(1) and (2) are equivalent Fischer projections.

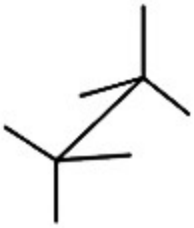
**SAWHORSE AND NEWMAN PROJECTION FORMULAE**

**Sawhorse projection formula :**

i) 

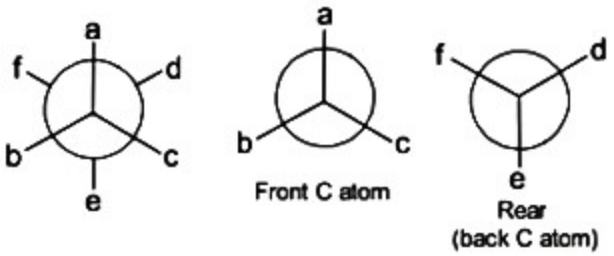
ii) 

iii) 

iv) 

**Newman Projection Formula :**

These projection formulae are obtained by viewing the molecule along the bond joining two carbon atoms.



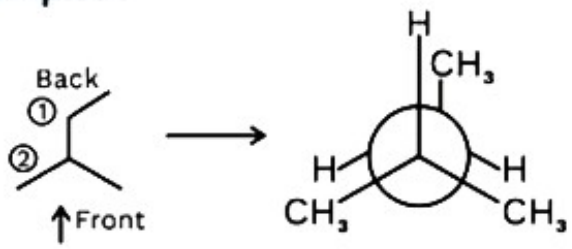
**Newman Projection :**

i) Front carbon as dot (.)

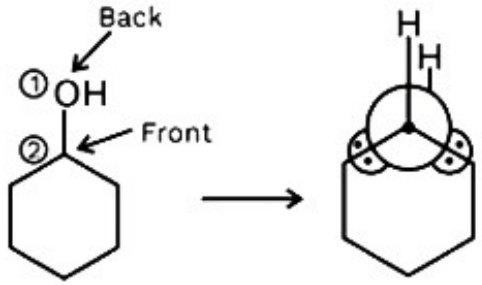
ii) Back carbon as circle 'O'

**Examples :**

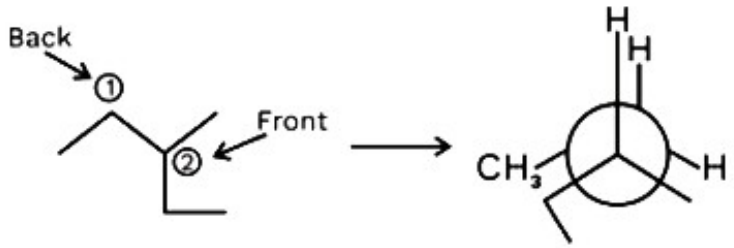
1.




2.




3.

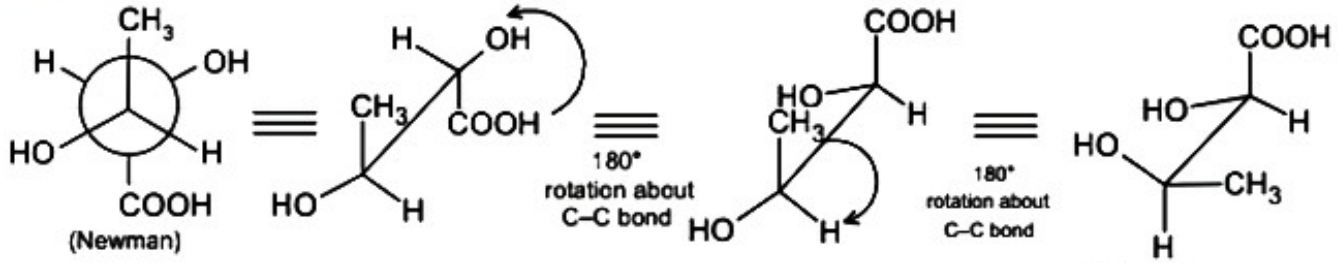



**Points to remember**

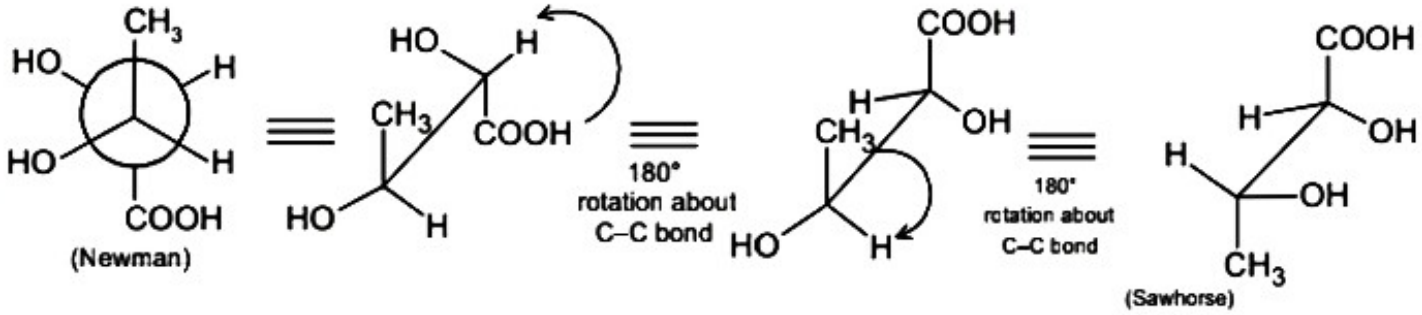
- Most of the compound could be represented in newman projection formula if particular atoms about which projection has to be made is mentioned.

**CONVERSION OF SAWHORSE AND NEWMAN PROJECTION FORMULAE IN FISCHER AND VICE VERSA :**

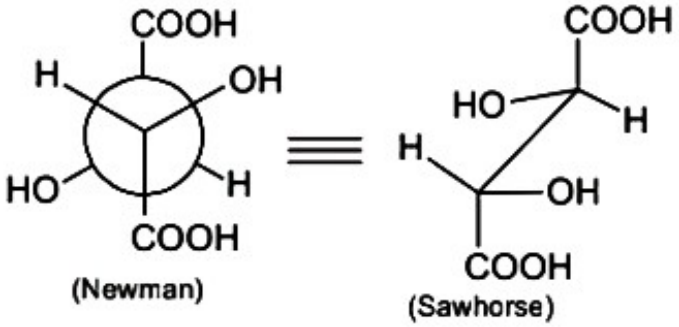
1.




2.

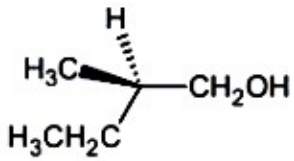


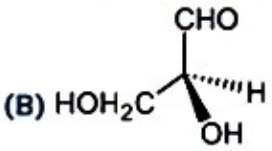

3.

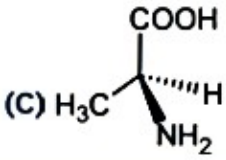


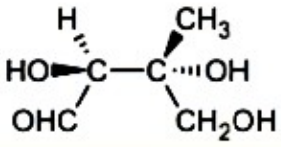
## ILLUSTRATIONS

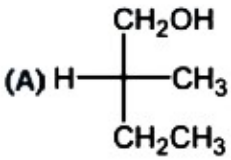
**Q1** **Draw Fischer projection formulae for the following molecules :**

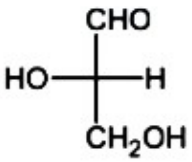
(A) 

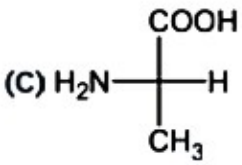

(B) 

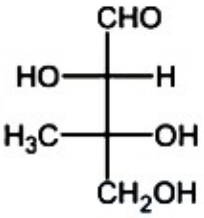

(C) 


(D) 

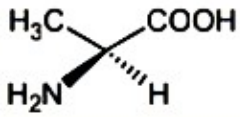

**Sol** (A) 

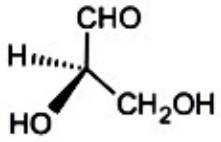

(B) 

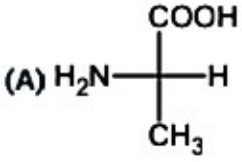

(C) 

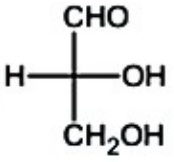

(D) 


**Q2** **Convert wedge formula to Fischer formula :**

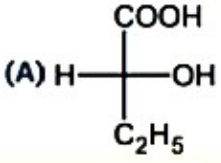
(A) 

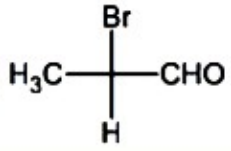

(B) 


**Sol** (A) 

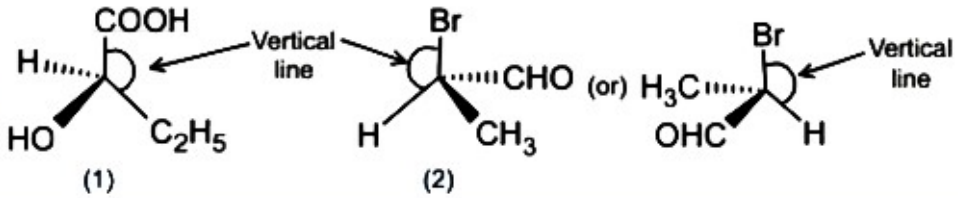

(B) 


**Q3** **Convert Fischer formula to wedge formula :**

(A) 


(B) 

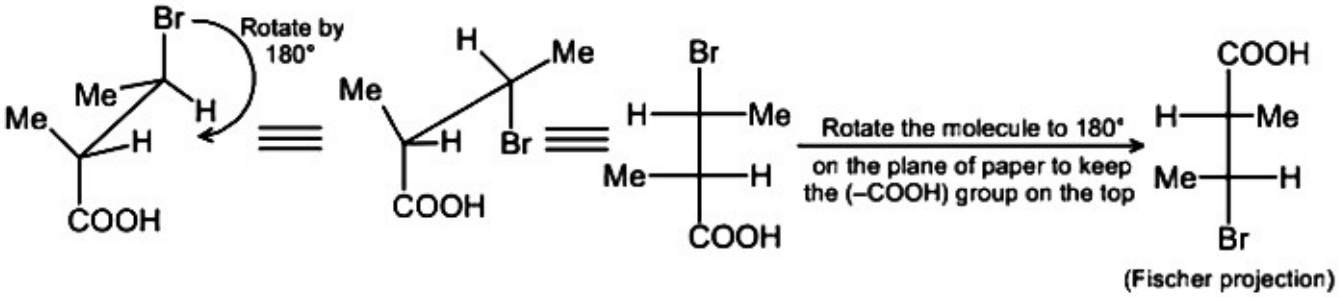
**Sol**

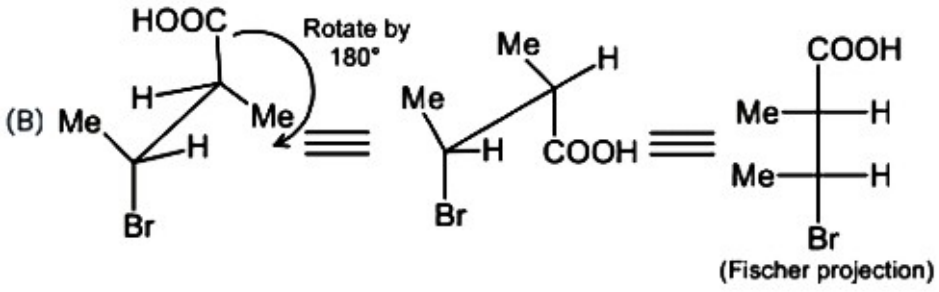



**Q4** **Convert Sawhorse formula to Fischer formula :**




**Sol** **(A)**



# Conformer and Conformation

## CONFORMER

There are ∞ conformers across given s bond.

Energy needed for C-C bond rotation is 3 Kcal per mole in ethane.

Conformers are freely interconvertable into each other and not separable, they are not true isomers.

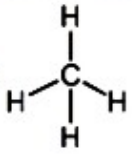
Conformers are present in dynamic equilibrium.

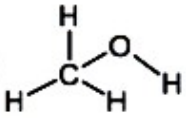
During conformational isomerism bond angle, bond length and configuration of molecule are not changed.

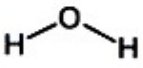
**Definitions**

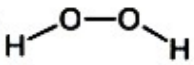
- Isomers formed by rotation across σ bond are called conformers.

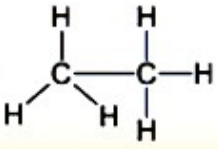
**Q1** **Which of the following shows conformational isomerism :**

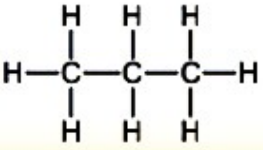
(A) 

(B) 

(C) 

(D) 

(E) 

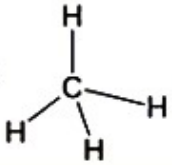
(F) 

**A1** **(B), (D) & (F)**

## BOND ANGLE (θ)

It is an angle between two valency of same atom.

**Points to remember**

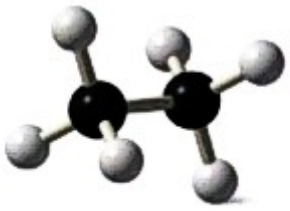
- Methane molecule ($CH_4$)  has bond angle between 2 C-H bonds is 109°28'.

## DIHEDRAL ANGLE (ϕ)

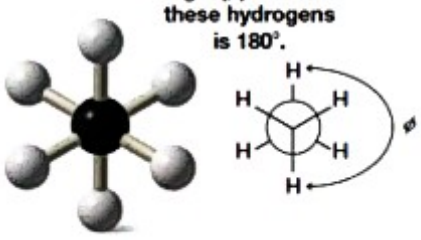
Dihedral angle is not fixed, it keep on changing at room temperature, that's why it is called rotational angle of conformer during conformation.

**Definitions**

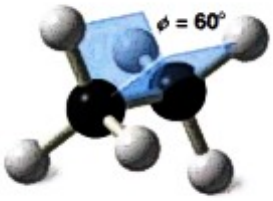
- The angle between two valency at different atom obtained in sigma bond rotation, is called dihedral ring.

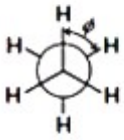


(a)



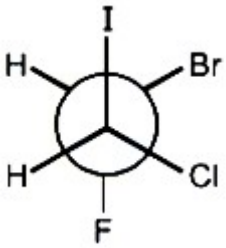

(b)




(c)

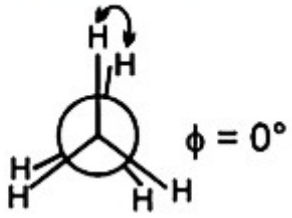
**Points to remember**




$\angle F - C - C - I = 180°$

$\angle F - C - C - Cl = 60°$

**ECLIPSED FORM**




Eclipsed conformation of ethane

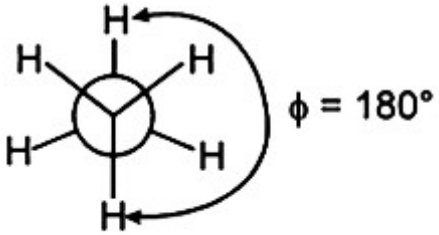
**Introduction**

When value of $\phi = 0°, 120°, 240°, 360°$ then, that form is known as eclipsed form in eclipsed form, valencies of one central atom completely overlapped by valencies of adjacent atoms.

**STAGGERED FORM**

When value of $\phi = 60°, 180°, 300°$ then that form is known as staggered form.

In staggered form, two biggest sized valencies attached at central atom must be opposite to each other.




staggered conformation of ethane

**SKEW FORM**

**Analysis of conformation :**

| | |
|---|---|
| Interelectronic Repulsion | EC > ST |
| Interelectronic Distance | ST > EC |
| Stability | ST > EC |
| Energy | EC > ST |
| Mole Fraction | ST > EC |

**Let's understand**

Infinite forms which are present in between eclipsed & staggered are known as skew forms.

$0° < \phi < 60°$

**MAJOR FACTORS TO BE CONSIDERED WHEN DEALING WITH CONFORMATIONAL ISOMERISM**

**Torsional energy (strain) :**

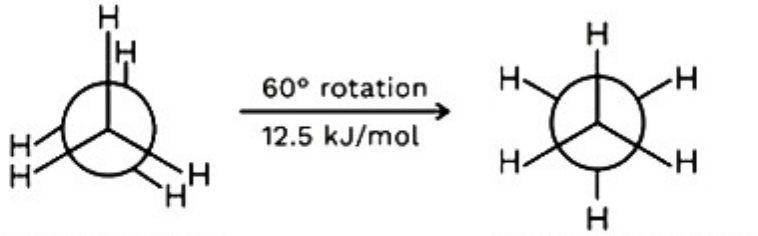
- Bond Pair-Bond Pair electronic repulsion.
- It is only considered in eclipsed conformer.

**Van der waal strain :**

- Steric strain or strain due to their size is Van der Waals repulsion.
- It is maximum in eclipsed conformer.

**Rotation Barrier :**

- Energy difference between most stable conformer & least stable conformer is known as rotation barrier.

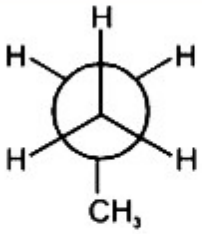



Eclipsed form Staggered form

**ILLUSTRATIONS :** Draw most stable structure.

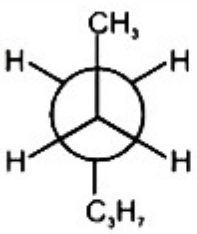
**Q1** **Propane**

**Sol**



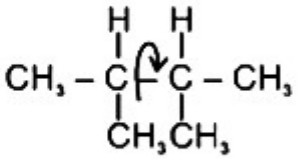
**Q2** **n-hexane ($C_2$ - $C_3$)**

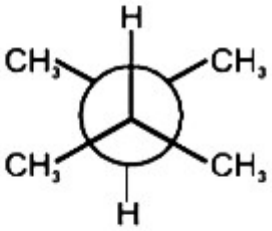
**Sol**



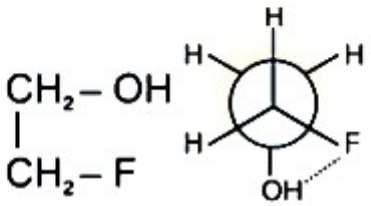
**Q3** **2-3 dimethyl butane ($C_2 - C_3$)**

**Sol**





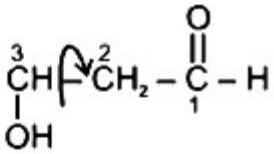
**Q4**



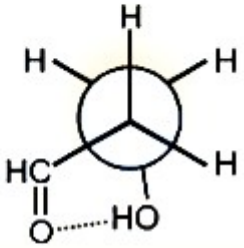

**Sol** Stability: Gauche > Anti

**Q5** **3-hydroxy propane ($C_2 - C_3$)**

**Sol**



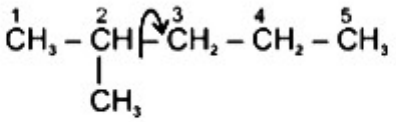
**Q6**

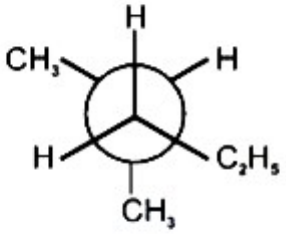


**Sol** (H-bonding) Gauche > Anti

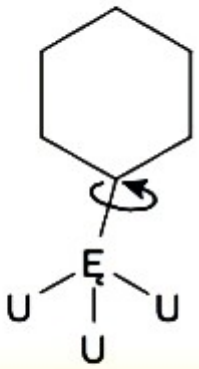
**Q7** **2-methyl pentane ($C_2 - C_3$)**

**Sol**

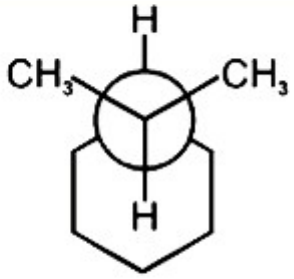




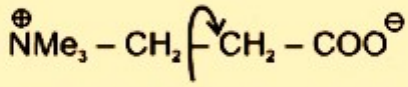
**Q8**



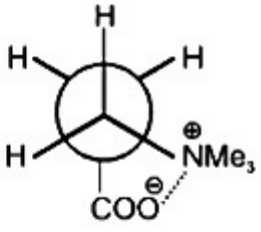
**Sol**



**Q9**



**Sol**



Gauche > Anti

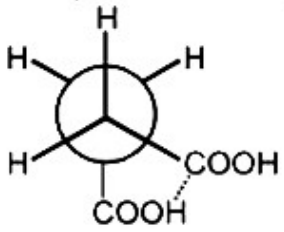
**Q10** $CH_2-COOH$
$|$
$CH_2-COOH$

**Sol** **(a)** Low pH (most stable)

**(b)** High pH (stable)

**Case-(a)**

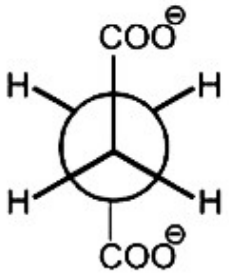
Low pH⇒ Solution (Acidic)




**Case-(b)**

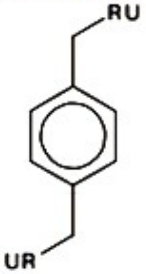
High pH⇒ Solution (Basic)

$COOH \rightarrow H^{\oplus} \uparrow \rightarrow COO^{\ominus}$

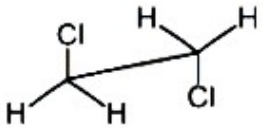



**Q11** **Among the given compound/s which having zero dipole moment in any of coformation?**

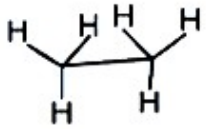
**(A)**




**(B)**

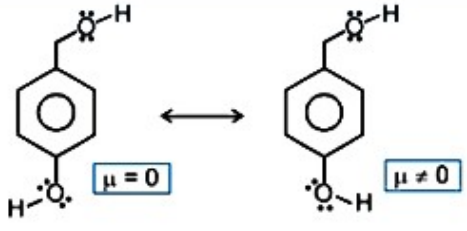



**(C)**

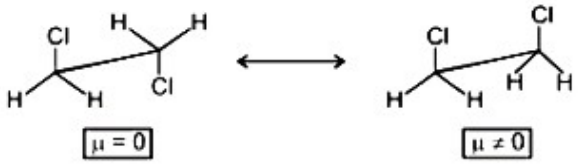



**(D) None of these**

**Sol** **(A)**




**(B)**




**(C)** Always same $\mu = 0$ $\therefore$ Only (C)

# Conformational Analysis in Cyclic Compound

## BAEYER ANGLE STRAIN THEORY

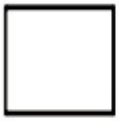
$$\text{Angle strain} \propto \frac{1}{\text{Stabilbity}} \propto \text{Reactivity}$$

Stability order of ring (cycloalkane)

6 > 5 ≃ 7 > 4 > 3

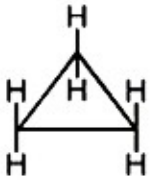
### Introduction

- According to Baeyer cyclic compounds are less stable then corresponding open chain compunds because in cyclic compound angle strain is present.

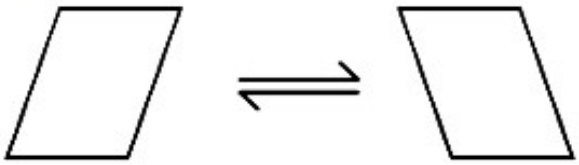
| COMPOUND | EXPECTED B.A. | REAL B.A. | ANGLE STRAINS |
|---|---|---|---|
|  | 109°28' | 60° | 49°28' |
|  | 109°28' | 90° | 19°28' |
|  | 109°28' | 108° | 1°28' |
|  | 109°28' | 109° | ≈ 0 |

**Cyclopropane**

- Present in forced eclipsed confirmation.
- Highly reactive due to its high angle strain.
- Highly reactive, therefore gives test of unsaturation.

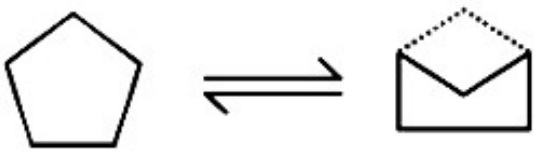


**Cyclobutane**



It undergo jelly flipping.

**Cyclopentane**



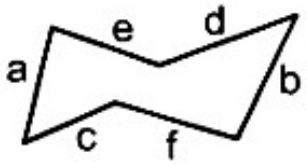
Envelope flipping

**Cyclohexane**

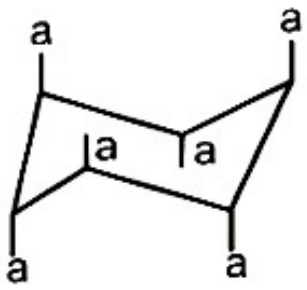
**Introduction**

- In real sense, cyclohexane is non-planar structure with negligible angle strain so that it is less reactive in nature & highly stable in nature.
- It's negligible angle strain also releases by various conformation.

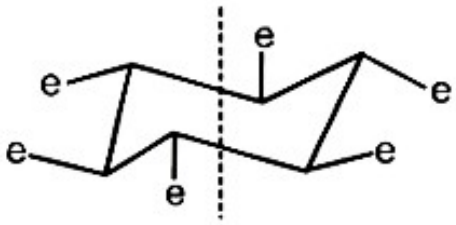
**CONFORMER OF CYCLOHEXANE**

1. 


a || b
e || f
c || d

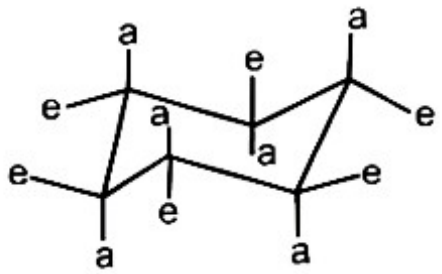
2. 


a → axial

3. 

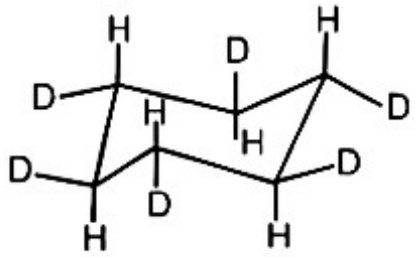

e → equatorial

4.



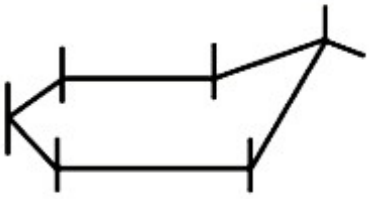

a & e → axial & equatorial

5.

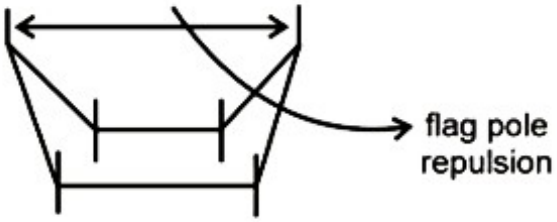



**Possible conformer of cyclohexane :**

a)



b)




c)

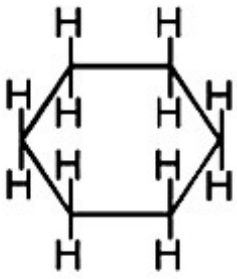


Twist boat

d)



e)

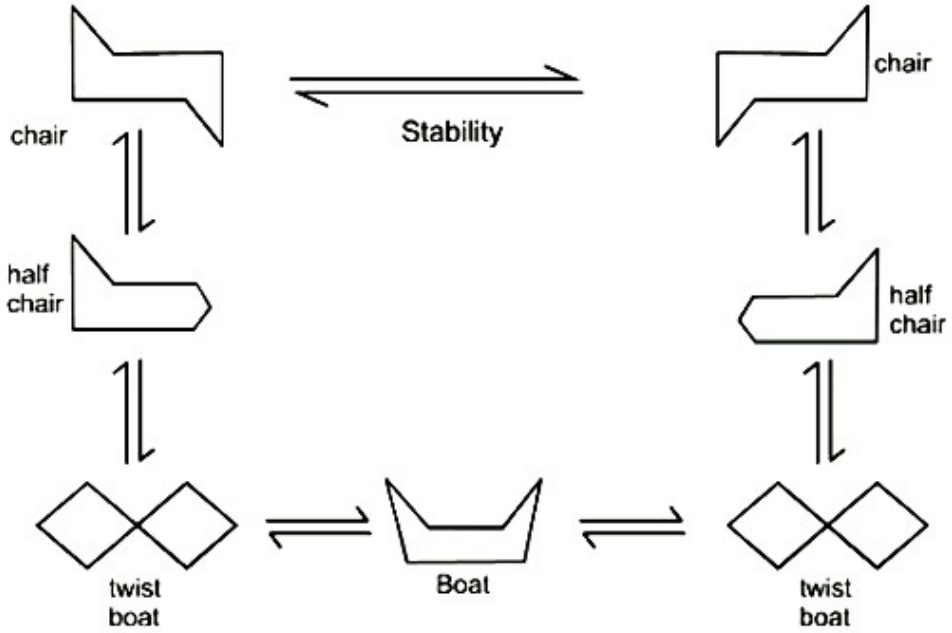



**Points to remember**

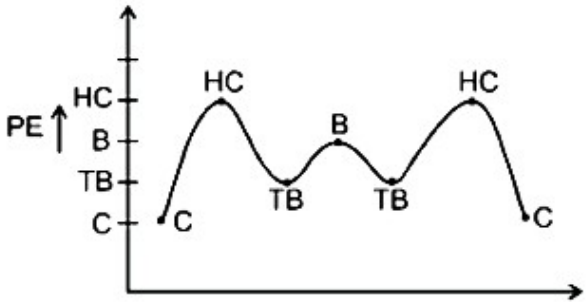
- Half chair form is very unstable due to eclipsing of hydrogen.
- Boat form is unstable due to flag pole repulsion.
- Twist boat is relatively less stable than chair form due to strain.
- Chair form has negligible angle strain.

  Therefore, most stable conformer of cyclohexane
- Rigid form does not exist because of eclipsing of twelve hydrogen.

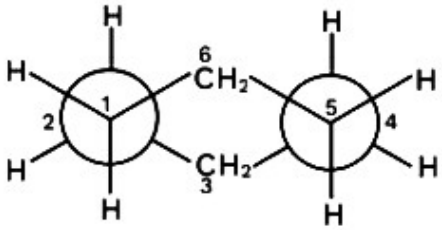
**Flipping**




During flipping, all equitorial bond will convert into axial & all axial bond will convert into equitorial bonds.

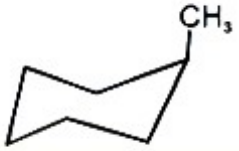



**Newman Projection of Cyclohexane Conformer (chair)**



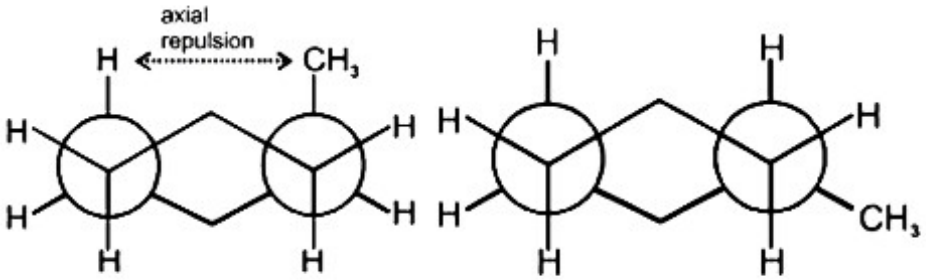

Newmann projection of the pair conformer

**Q1** **Which conformer of methyl cyclohexane is more stable?**

(A) 


(B) 


**Sol**

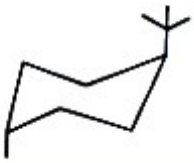



- Equitorial is more stable than axial.
- Bulky group must occupy equitorial position.
- (A) is unstable because in (A), 1,3 diaxial repulsion is present

**Q2** **Compare relative stability :**

(A) 

(B) 

(C) 

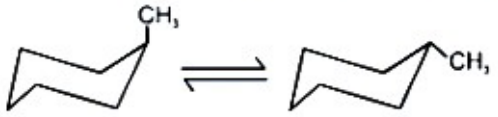
(D) 

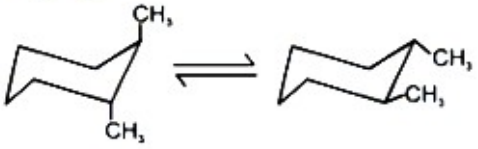
**Sol**

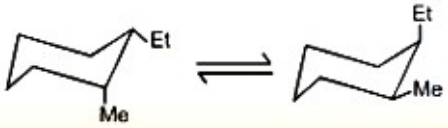
* Chair
* Twist boat
* Boat
* Half chair
* Rigid

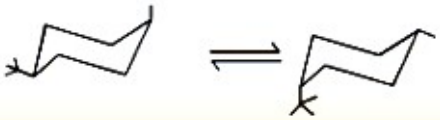


**Q3** **Find the direction of equilibrium and also identify $K_{eq}$.**

**(A)** 


**(B)** 


**(C)** 

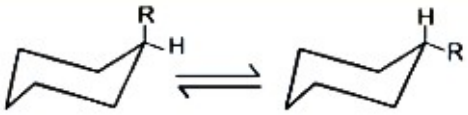

**(D)** 

**Sol** **(A)** forward, $K_{eq} > 1$

**(B)** forward, $K_{eq} > 1$

**(C)** backward, $K_{eq} < 1$

**(D)** backward, $K_{eq} < 1$

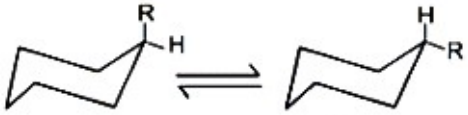
**Q4** 


**Explain why substitution at equatorial position changes $K_{eq}$ value drastically up.**

**Sol**

| **–R** | **$K_{eq}$** | **–R** | **$K_{eq}$** |
|---|---|---|---|
| –H | 1 | $-CH_3$ | 18 |
|  | 35 |  | 4000 |

Bulky group is more stable at equatorial as bulkiness increases stability at equatorial position increases.

**Q5** 


| **–R** | **$K_{eq}$** |
|---|---|
| **–F** | **1.5** |
| **–Cl** | **2.4** |
| **–Br** | **2.2** |
| **–I** | **2.2** |

**Explain?**

**Sol** In case of F and Cl, size factor dominates therefore more bulky group at equatorial is more stable.

But in case of Cl, Br and I bond length factor dominates.

# Geometrical Isomerism

## GEOMETRICAL ISOMERISM

Those stereoisomers which are not interconvertible in each other at normal temperature are called configrational isomers.

Configurational isomers which are not interconvertible in each other at normal temperature due to restricted rotation are called geometrical isomers.

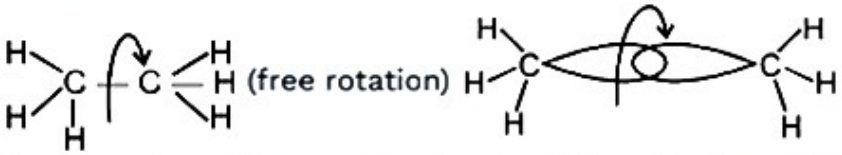
**Definitions**

- Those stereoisomers which have different geometry around restricted rotation unit.

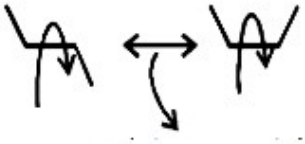
### Three Condition for Geometrical Isomers

- Restricted rotation
- Different axial distance
- Different unit at terminal (terminal valency should be different)

### Free Rotation System

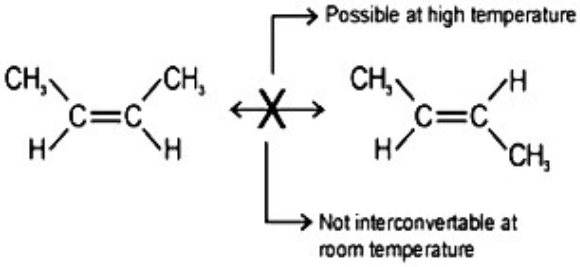


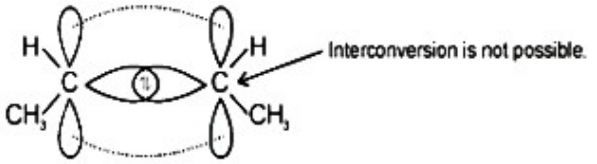

**For example :** Butane molecule about $C_2 - C_3$ sigma bond shows rotation.



Interconvertible at room temperature

### 1. Restricted Rotatory System :




In a molecule restricted rotation arises due to

**a)** Presence of double and triple bond



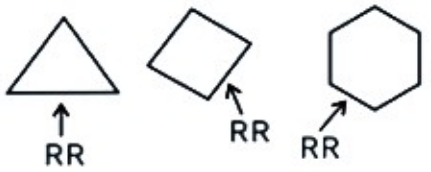

**Points to remember**

- Triple bond never show G.I.

**Example :** $CH_3-C\equiv C-CH_3$ ($\ell_1 = \ell_2$)

**b)** Around single bond in cyclic compounds (σ bond pure)

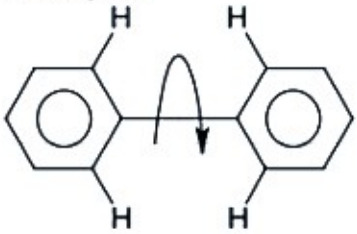
DU = 1

**Example :** 

act like double bond

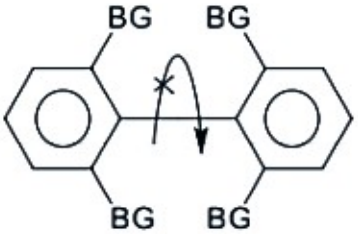
**c)** Polyphenyl having Steric Hindrance

**Example :**




Free rotation

- Small group
- Steric hindrance decreases
- Rotation (free)

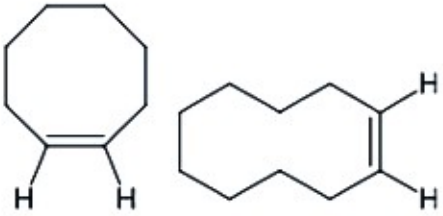



BG = Bulky group

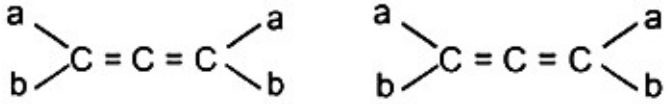
- Bulky group
- Steric hindrance increases
- Rotation (restricted)

[*This is because of steric hindrance rings become perpendicular so, that they don't come back.*]

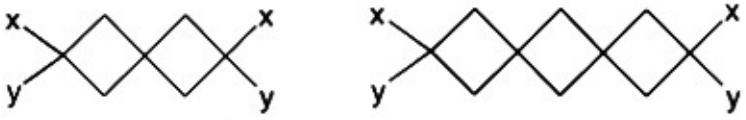
**d)** Cyclic alkene




**e)** Allene System

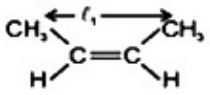
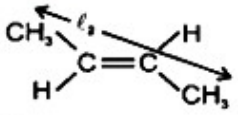



**f)** Spirane System



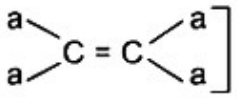

We will study this d, e, f cases in next lecture.

**2. Different aerial distance :**

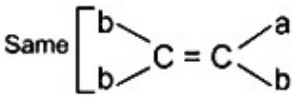



Here $\ell_1 \neq \ell_2$

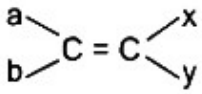
**3. Different Terminal Valency at same atom :**

**a)** 

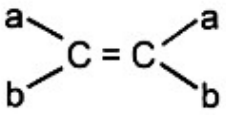
RR✓
TVD✖
$\ell_1 \neq \ell_2$ ✖
GI✖

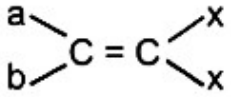
**b)** 

RR✓
TVD✖
$\ell_1 \neq \ell_2$ ✖
GI✖

**c)** 

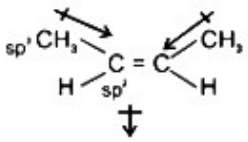
RR✓
TVD✓
$\ell_1 \neq \ell_2$ ✓
GI✓

**d)** 

**e)** 

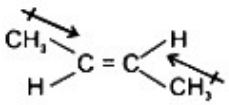
**SPECIAL POINT TO LEARN**

**Example :** Dipole moment and Boiling Point :



$\mu \neq 0$

(A)



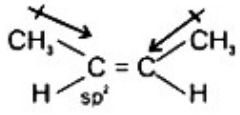
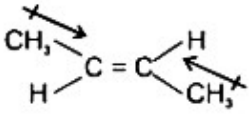
$\mu = 0$

(B)

BP $\propto \mu$ (Dipole moment)

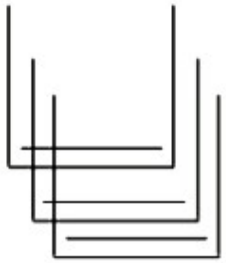
$BP_A > BP_B$

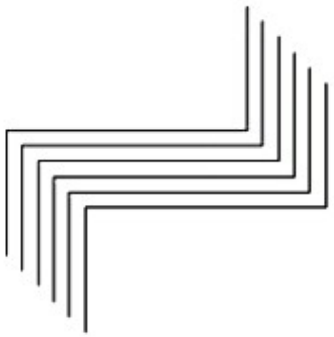
**Example :** Melting Point

**Points to remember**

- Geometrical isomerism have different physical & chemical properties





Efficient packaging ✗ $MP_A < MP_B$ Efficient packaging ✓

(MP)↑ (MP)↓

**Geometrical isomerism in cyclic alkane compound**

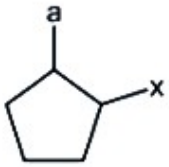
**Examples :** Select among the given compounds, which is/are capable to show geometrical isomerism.

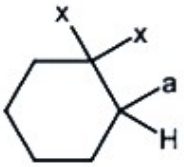
**Introduction**

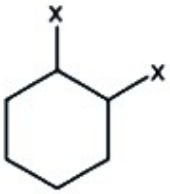
- In cycloalkane, if two different $sp^3$ hybridised atom contain different groups then molecule show Geometrical isomerism.

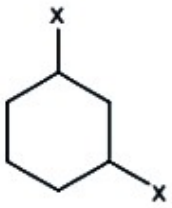
**1.**  No

**2.**  Yes

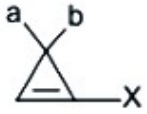
**3.**  Yes

**4.**  No

**5.**  Yes

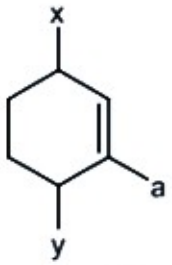
**6.**  Yes

**7.**  Yes

**8.**  No

**9.**  No

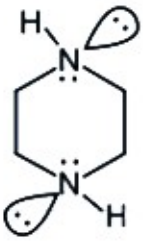
**10.**  Yes

11. 

Yes

12. 

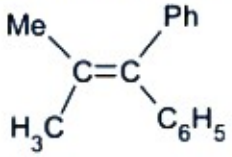
Yes

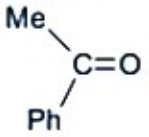
13. 

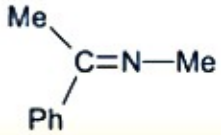
Yes

**ILLUSTRATION**

**Q1** **Which of the following double bond will not exhibit geometrical isomerism.**

**(A)** 

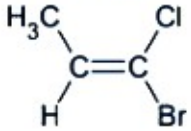

**(B)** 

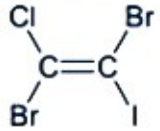

**(C)** 

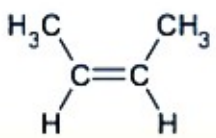

**(D)** Me – N = N – Me

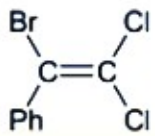
**A1** **(A, B)**

**Q2** **Which of following will not show geometrical isomerism ?**

**(A)** 

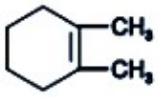

**(B)** 


**(C)** 

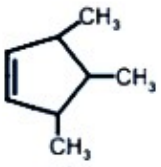

**(D)** 


**A2** **(D)**

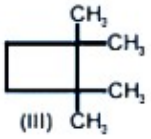
**Q3**




(I)




(II)




(III)

**Which among these are isomers ?**

**(A)** I and II

**(B)** I and III

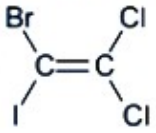
**(C)** II and III

**(D)** all of these

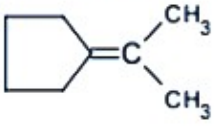
**A3** **(A)**

**Q4** **Which of the following compound can show geometrical isomerism.**

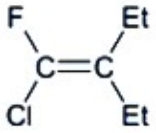
**(A)**



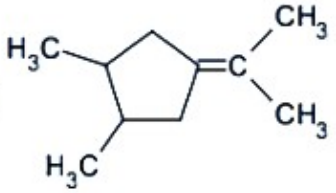

**(B)**




**(C)**



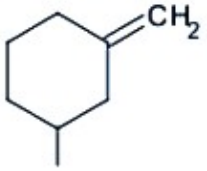

**(D)**



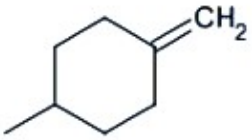

**A4** **(D)**

**Q5** **The geometrical isomerism is shown by**

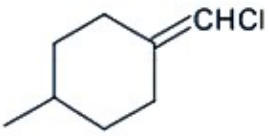
**(A)**



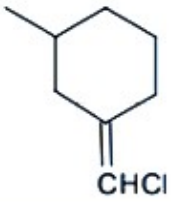

**(B)**




**(C)**




**(D)**




**A5** **(D)**

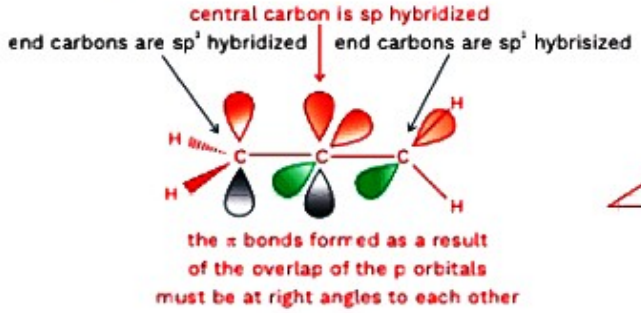
# Poly Alkene

**CUMULATED ALKENE** $CH_2 = C = C = C = C = CH_2$

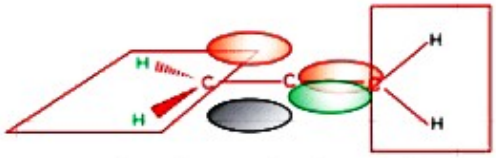
**CONJUGATED ALKENE** $CH_2 = CH - CH = C - CH = CH_2$

**ISOLATED ALKENE** $CH_2 = CH - CH_2 - CH = CH - CH_2 - CH = CH_2$

**Stereochemistry in even p-bond allene:**

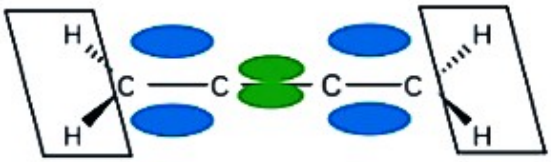



the π bonds formed as a result of the overlap of the p orbitals must be at right angles to each other



not only are the two π bonds perpendicular, but the two methylene groups are too

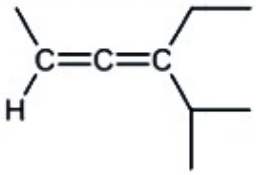
**Stereochemistry in Odd π-bond allene :**

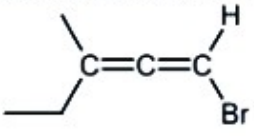


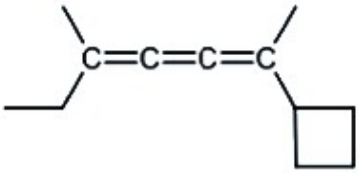
**Both terminal sides valencies are in same plane**

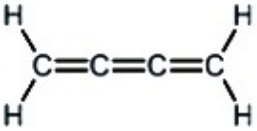
**Condition for cumulated alkene :**

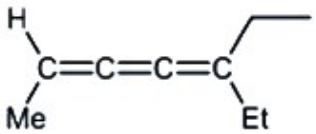
- Even cumulated alkene here show geometrical isomerism
- Odd cumulated alkene can show geometrical isomerism if terminal valencies are different.

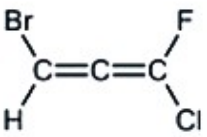
i) 

ii) 

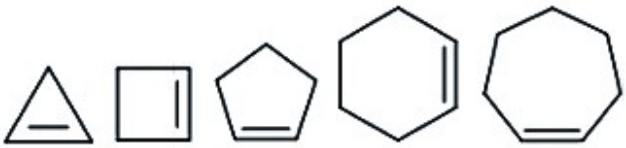
iii) 

iv) 

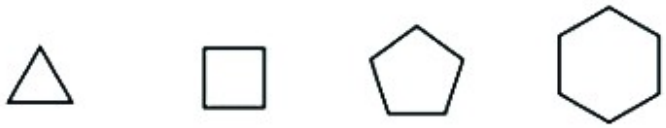
v) 

vi) 

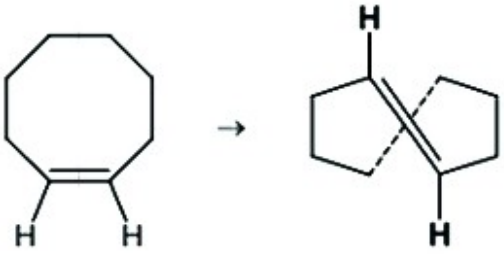
**GI IN CYCLO ALKENE**



Baeyer angle strain theory :



| 109°28' | 109°28' | 109°28' | ≈0° |
|---|---|---|---|
| 60° / 49°28' | 90° / 19°28' | 108° / 1°28' | |



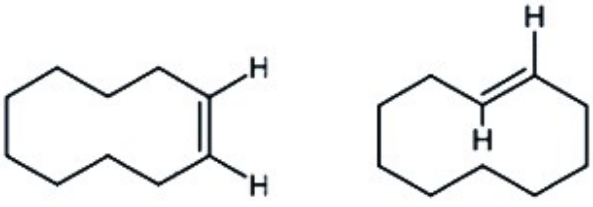

cis-cyclooctene trans cyclooctene

stability

8 to 11 ⟶ cis > trans

11 ⟶ trans > cis

**Example :**




cis-decene trans-decene

**Points to remember**

- They can not show geometrical isomerism about alkene due to angle strain in trans form.

**Q1** **Minimum number of carbon required for cycloalkene to show geometrical isomerism**

**A1** 6

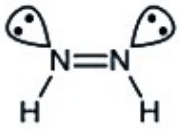
**Sol**



**Q2** **Minimum number of carbon required to show geometrical isomerism.**

**A2** 0

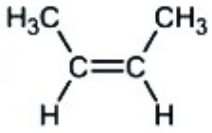
**Sol**




**Q3** **Minimum number of carbon required for alkene to show geometrical isomerism**

**A3** 4

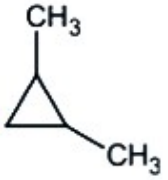
**Sol**




**Q4** **Minimum number of carbon required for cycloalkane to show geometrical isomerism**

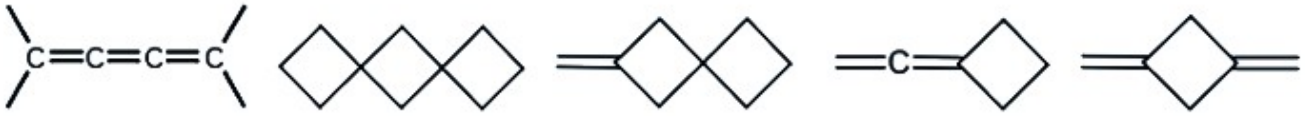
**Q4** 5

**Sol**

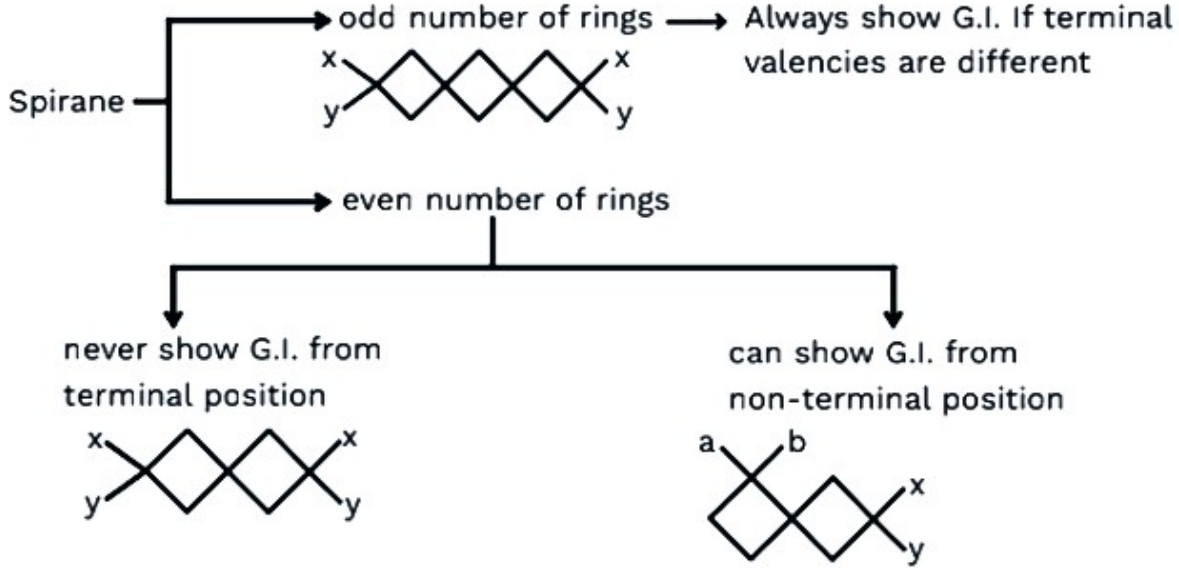



**GEOMETRICAL ISOMERISM IN SPIRANE**

By replacing ring from π-bond or π-bond from ring basic geometry remain same.

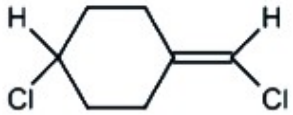


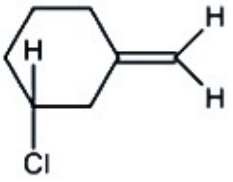

Above all have same geometry.

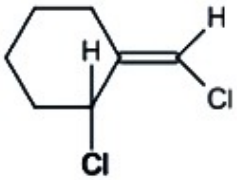


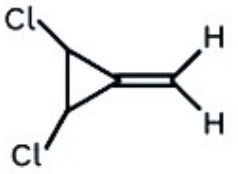
1. 

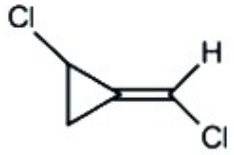
2. 

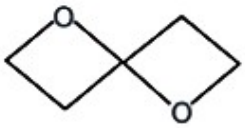
3. 

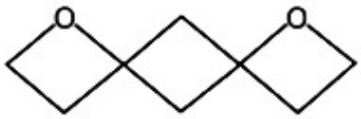
4. 

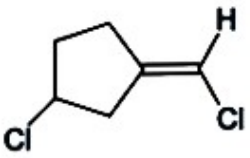
5. 

6. 

7. 

8. 

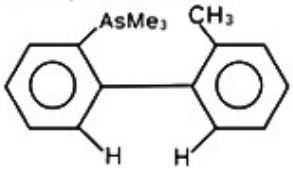
9. 

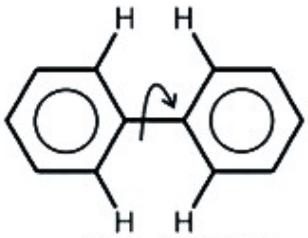
10. 

11. 

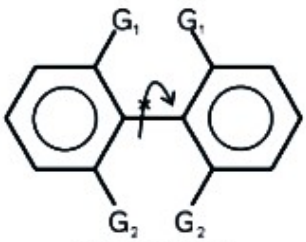
## GEOMETRICAL ISOMERISM IN POLYPHENYL

**Atropisomers :**




Free rotation



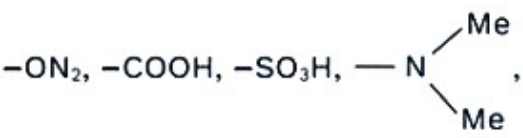
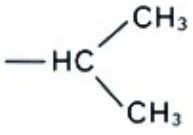

Restricted rotation

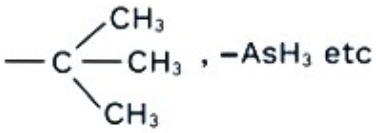
$G_1 - G_2 \Rightarrow$ bulky groups at ortho position

**Definitions**

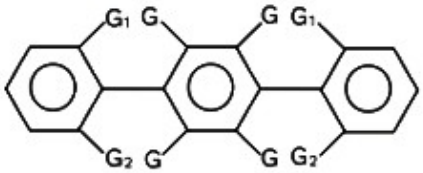
- Stereoisomers arises due to bulky group or stereo repulsion are called atropisomers.

**Points to remember**

- Small groups : –H, –D, –T, –OH, $-NH_2$, –F, –Cl, –CH–, –C ≡ C–, –C ≡N etc.
- Bulky groups : –Br, –I, $-ON_2$, –COOH, $-SO_3H$,  , $CH_3$ ,  ,

 , $-AsH_3$ etc

**Example:**




Restricted rotation unit ✓
Terminal valency different ✓
geometrical isomerism ✓

**Note :**

Even phenyl system never show geometrical isomerism

Odd phenyl system can show geometrical isomerism If terminal valencies are different at ortho position.

# Gometrical Isomerism

**TYPES OF CONFIGURATION IN GEOMETRICAL ISOMERISM**

C is trans

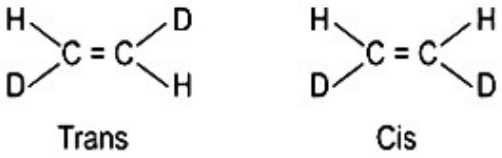
E – Z

Syn – Anti

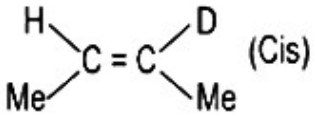
**1. CIS-TRANS**

**Example :**

**a)**

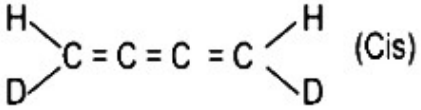



**b)**



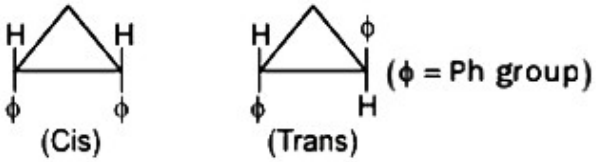

(Cis)

**c)**



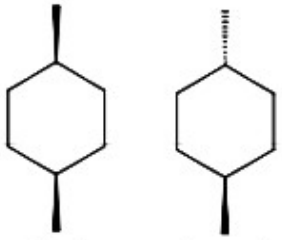

(Cis)

**d)**

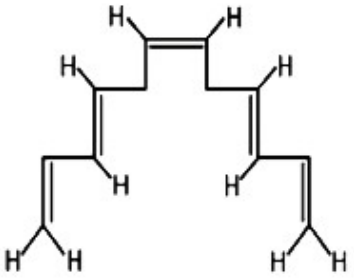



(ϕ = Ph group)

**e)**



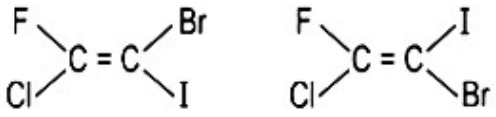

**f)**



**g)**



**h)**



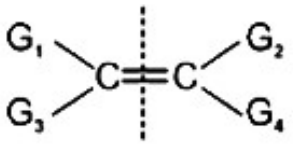

**Introduction**

- Same group oriented to same side at terminal valencies- **CIS**
- Same group oriented to different side at terminal valencies- **TRANS**

## 2. E-Z SYSTEM

E = Entgegen & Z = Zusamman

- There is no relationship between E-Z & cis-trans means a molecule having E-Z configuration, may be cis or trans & vice versa.

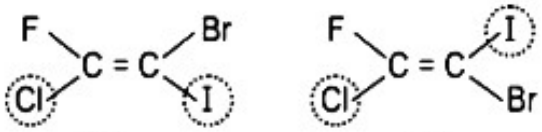


**Points to remember**

- Same priority group are on same side ⇒ Z
- Same priority group are on opposite side ⇒ E

**Priority order or rule : (Cahn, Ingold & Prelog ,CIP rule)**

**Rule-1 :** Assign priority on the basis of atomic number, higher is the atomic number of connecting atom from Geometrical Isomer unit higher will be its priority.

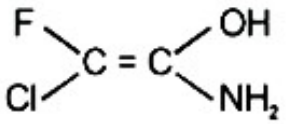


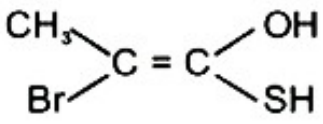
(Z) (E)

**Priority order :** $-I > -Br > -Cl > -F > -OH > -NH_2 > -CH_3$

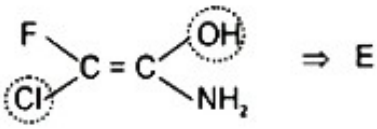
**Concept Building Exercise**

**Example :**

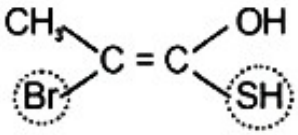




**Sol.**



⇒ E

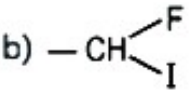
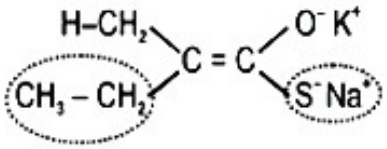


⇒ Z

**Rule-2 :** If first atom attached with terminal valencies is same consider second atom and if it is also same then see till first point of difference is not found.

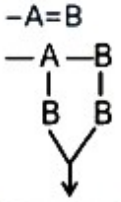
**Concept Building Exercise**

**Example :**

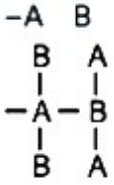
a) $-CH_2 - Cl > -CH_2 - CH_2 - I$ b)  $< -OH$ c) 

**Rule-3 :** If connecting valencies having multiple bonds then duplicate the double bond & triplicate the triple bond then check priority.

**Duplicate**

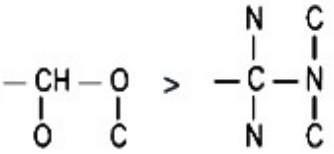



**Triplicate**

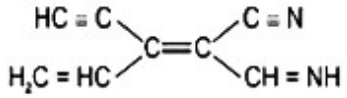



**Example :**

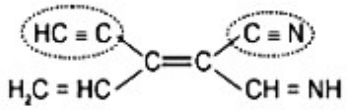
**1.** –CH=O–C≡N

**Sol.** 

**2.**




**Sol.**

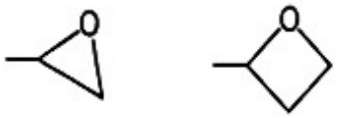



(Z)

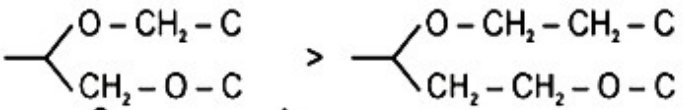
**Rule-4 :** If cyclic groups are present then priority will be decided by breaking single bond to obtain open chain compound.

**Examples :**

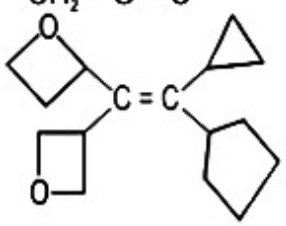
**1.**



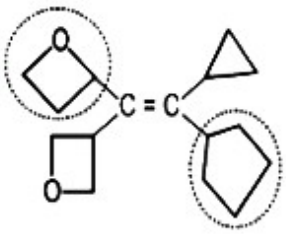

**Sol.**




**2.**

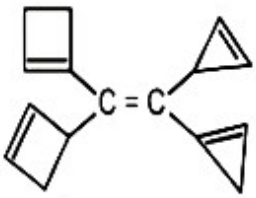



**Sol.**

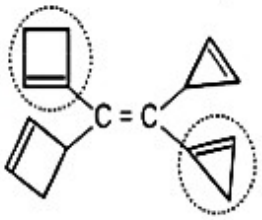



(E)

**3.**



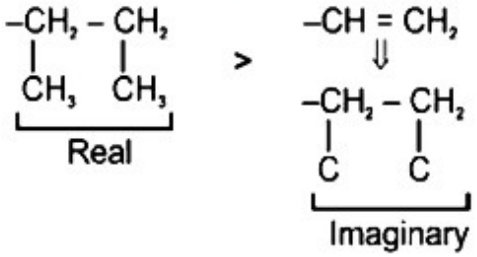

**Sol.**




(E)

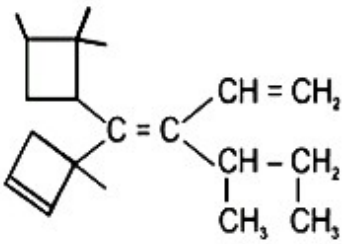
**Rule-5 :** If real group is exactly identical to hypothetical (imaginary) group then priority will be given to real group.

Real > Imaginary

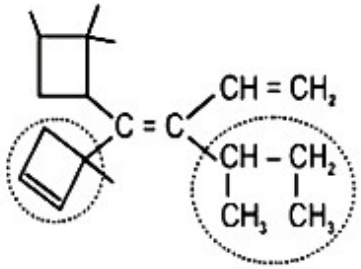



**Example :**

**1.**



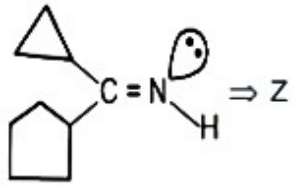

**Sol.**

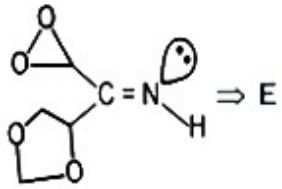



(Z)

**Rule-6 :** If lone pair is present then it will considered as least priority group as lone pair atomic number is zero.

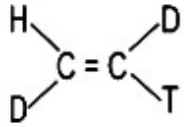
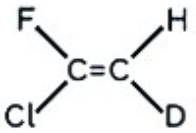
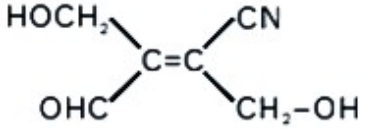
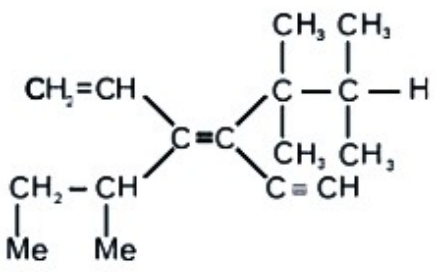
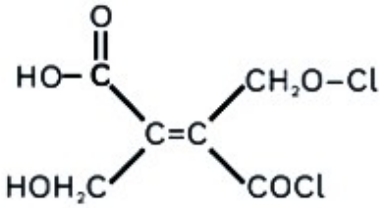
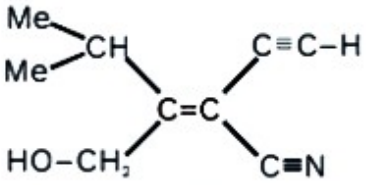
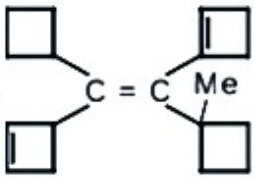
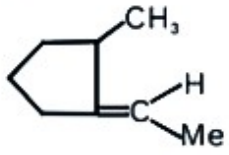
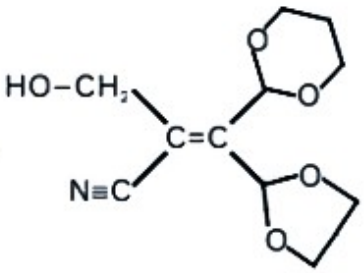
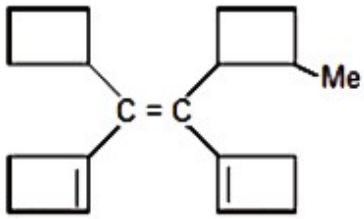
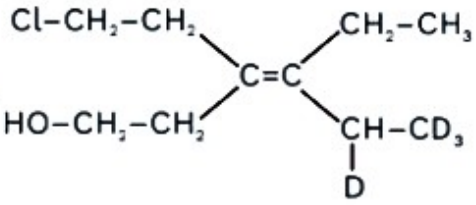
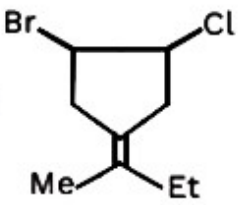
**Example :**

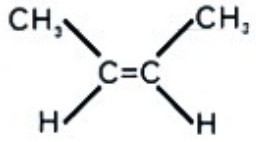
1.  ⇒ Z

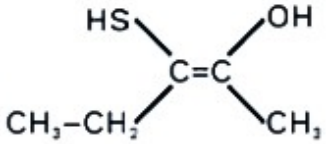
2.  ⇒ E

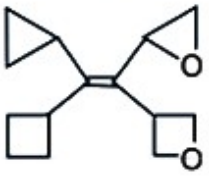
**Rule-7 :** In case of isotope, priorities will be decided according to atomic mass, higher will be atomic mass, higher is the priority.

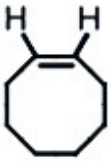
**Examples :**

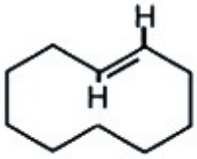
1. $-T > -D > -H$
2. $-OH < -O^{18}H$
3.  (trans) / (Z)
4.  (Z)
5.  (Z)
6.  (Z)
7.  (E)
8.  (E)
9.  (Z)
10.  (E)
11.  (E)
12.  (E)
13.  (Z)
14.  (E)
15.  (E)

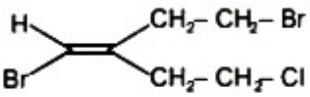
16.  (Z)

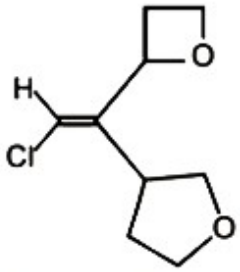
17.  (Z)

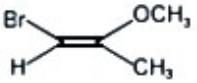
18.  (E)

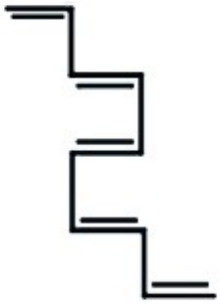
19.  (Z)

20.  (E)

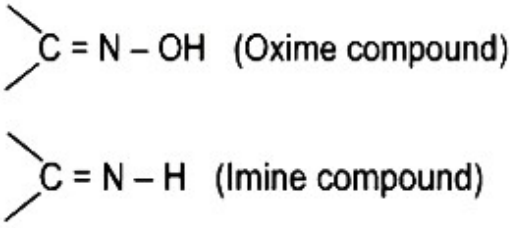
21.  (E)

22.  (E)

23.  (Z)

24.  (EEE)

### 3. SYN - ANTI

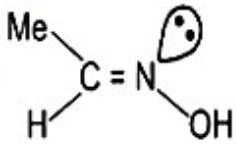


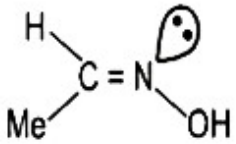
C = N – OH (Oxime compound)

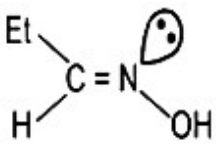
C = N – H (Imine compound)

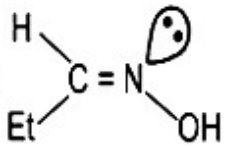
– N = N – (Azo compound)

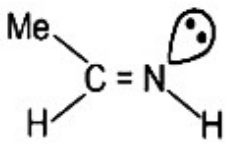
**Examples :**

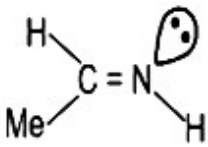
(1) 

(2) 

(3) 

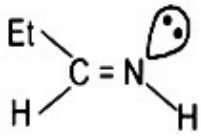
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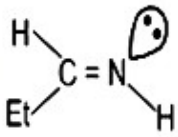
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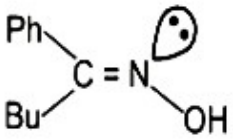
(6) 

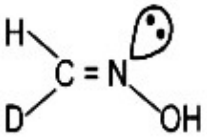
**Introduction**

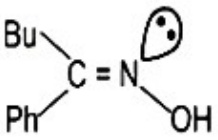
- Applicable only for nitrogen containing compound where geometrical isomerism is possible.

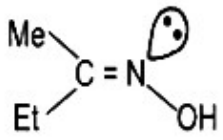
(7) 

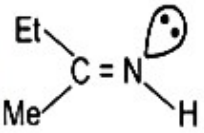
(8) 

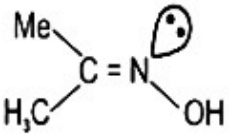
(9) 

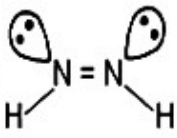
(10) 

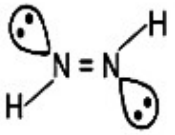
(11) 

(12) 

(13) 

(14) 

(15) 

(16) 

# Calculation of Geometrical Isomer

**CALCULATION OF GEOMETRICAL ISOMER**

Case I ⇒ If both the ends are different

$2^n$ when n is number of stereogenic area or Pi bond which can show G.I.

Case II ⇒ If both the ends are same

$2^{n-1} + 2^{p-1}$

If n = even ; P = n/2

If n = odd ; $P = \frac{n+1}{2}$

**Case-1 : Unsymmetrical compounds :**

**1.** $CH_3 - CH = CH - CH = CH - CH_2 - CH_3$

**Ans.** 4

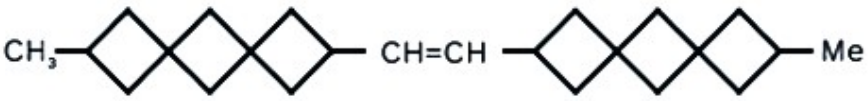
**2.** $CH_3-CH=CH-CH=CH-CH(CH_3)(CH_3)$

**Ans.** 4

**Case-2 : Symmetrical compounds :**

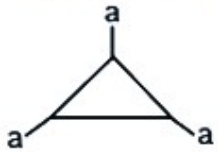
**3.** $CH_3 - CH_2 - CH = CH - CH = CH - CH_2 - Me$

**Ans.** 3

**4.** 


**Ans.** 6

**Case-3 : In case of cyclic compound draw and check.**

**5.** 


**Ans.** 2

**Sol.** 

**6.** 

**Points to remember**

**Rules for cyclic system :**

- 3 member to 7 member cyclo alkene exist in only cis form.
- 8 to 11 member can form cis & trans but cis is more stable
- from 12 member trans is more stable.

**Ans.** 3

**Sol.**

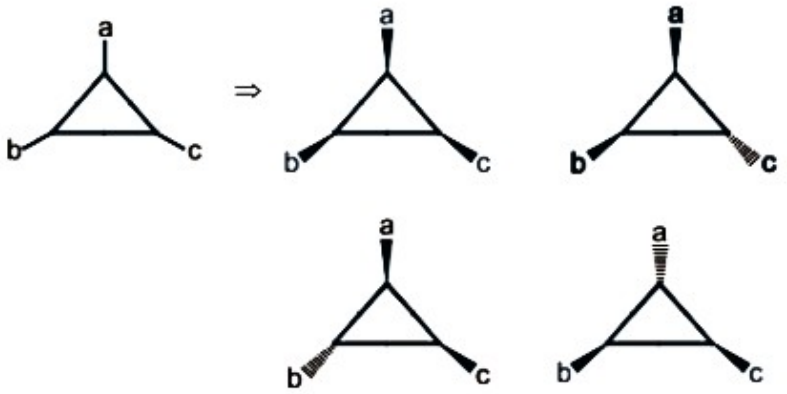


**7.**



**Ans.** 4

**Sol.**




**Case-4**

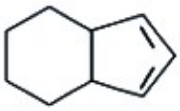
**8.** $CH_3-CH=C\begin{matrix}HC=CH-CH_3\\HC=CH-CH_3\end{matrix}$

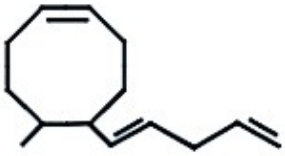
**Ans.** 4

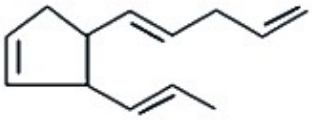
**9.** $CH_2=C\begin{matrix}CH=CH-CH_3\\CH=CH-CH_3\end{matrix}$

**Ans.** 3

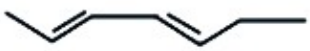
**Q1** **Calculate total number of only geometrical isomers in following compounds (Theoritical).**

(1)  Total G.I. = 2

(2)  Total G.I. = 8

(3)  Total G.I. = 8

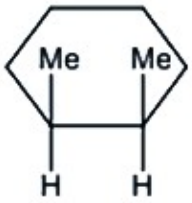
(4)  Total G.I. = 8

(5)  Total G.I. = 4

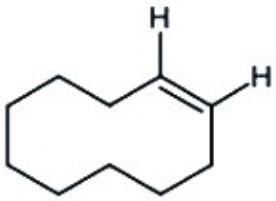
(6)  Total G.I. = 3

(7)  Total G.I. = 8

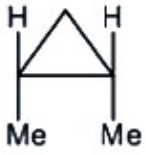
(8)  Total G.I. = 6

(9) 

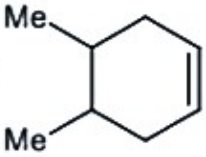
Total G.I. = 2

(10) 

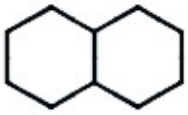
Total G.I. = 2

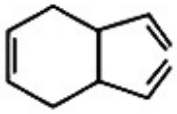
(11) 

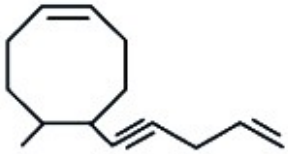
Total G.I. = 2

(12) 

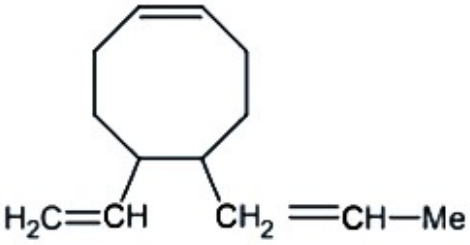
Total G.I. = 2

(13)  Total G.I. = 2

(14)  Total G.I. = 2

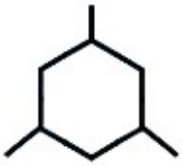
(15)  Total G.I. = 4

(16) $CH_3-CH=CH-CH=CH-CH=CH-CH_3$ Total G.I. = 6

(17)  Total G.I. = 8

(18) $Me-CH=C=CH-CH_2-CH=C=C=CH-Me$ Total G.I. = 2

(19) $Br-CH=C=C=CH-CH=CH-Br$ Total G.I. = 4

(20)  Total G.I. = 2

# Chiral Centre & | C.C.| OR Asymmetric - Centre OR Stereogenic - Centre

**Rule :-**

i) Consider atomic number for preference.

ii) If lone pair then consider is zero atomic number.

iii) If isotope then consider atomic mass for preference.

iv) If first valaney is same then consider second and so on.

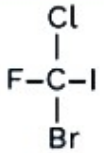
v) If p-bond then consider (s) sigma bond by adding hypothetical atom.

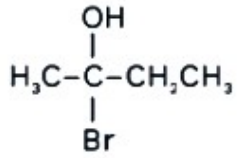
vi) If cyclic system then consider as open chain system.

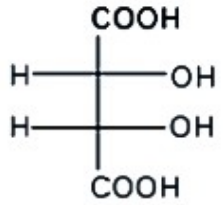
**Definitions**

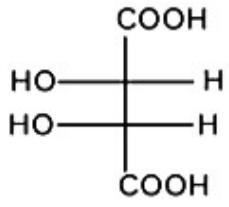
- Those $sp^3$ atom which is having four different valancy
- To identify chiral centre consider C.I.P. rule (E and Z)

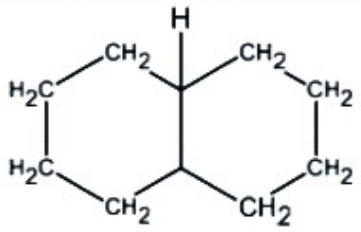
**Q1** **Calculation of chiral centre in below mention organic compound :**

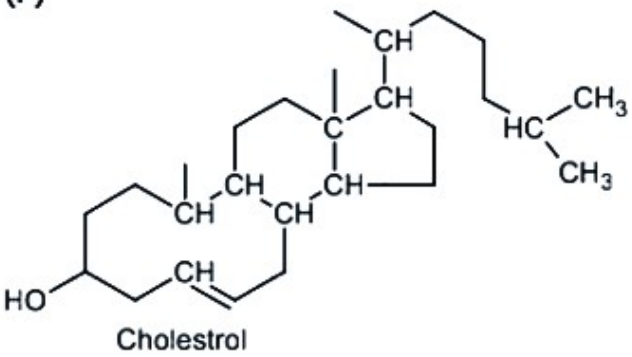
**(A)**  (1)

**(B)**  (1)

**(C)**  (2)

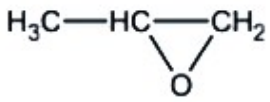
**(D)**  (2)

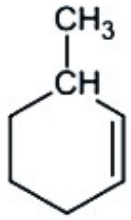
**(E)**  (0)

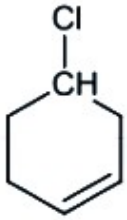
**(F)** 

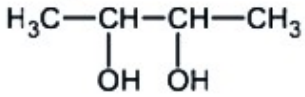
Cholestrol

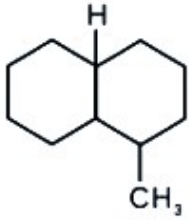
(9)

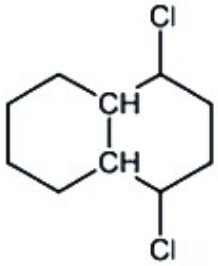
(G)  (1)

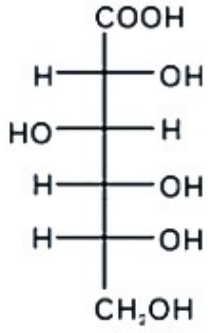
(H)  (1)

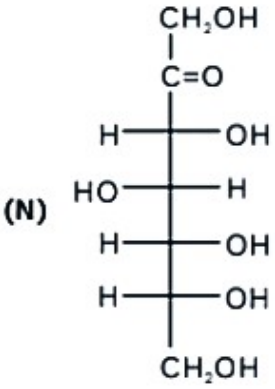
(I)  (1)

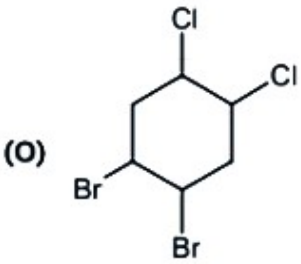
(J)  (2)

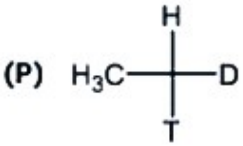
(K)  (2)

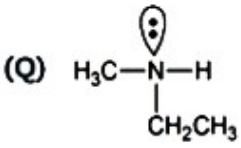
(L)  (4)

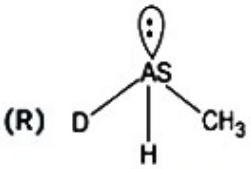
(M)  (4)

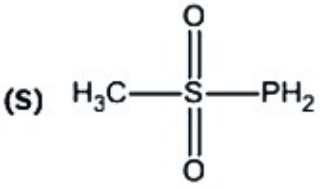
(N)  (4)

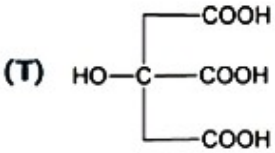
(O)  (4)

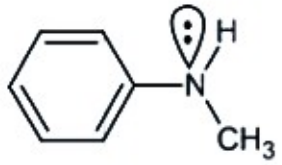
(P)  (1)

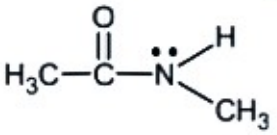
(Q)  (1)

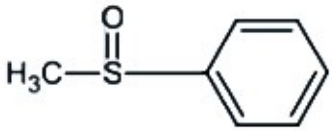
(R)  (1)

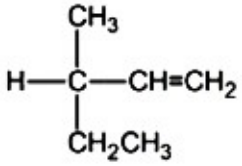
(S)  (1)

(T)  (1)

(U)  (1)

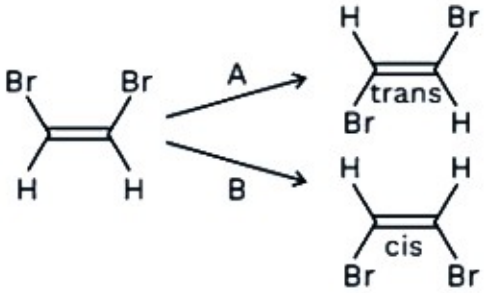
(V)  (1)

(W)  (1)

(X)  (1)

**STEREO-CENTRE / S.C**

by interchanging any two valancies and we get new stereoisomer.

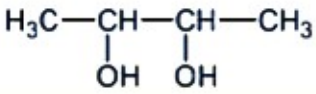



**Definitions**

- The specific centre in which one interchange give a new stereoisomer

**ILLUSTRATION**

**Q1** **Compound**



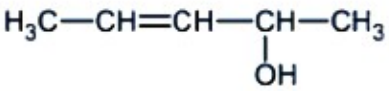
**A1**

**C.C.**

(2)

**S.C.**

(2)

**Q2** 

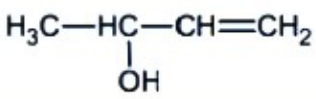
**A2**

| c.c. | s.c. |
|---|---|
| (1) | 1 + 2 = 3 |

**Q3** $CH_3-CH=CH-CH=CH-CH_3$

**A3**

| c.c. | s.c. |
|---|---|
| 0 | 4 |

**Q4** 

**A4**

| c.c. | s.c. |
|---|---|
| (1) | (1) |

**Q5** 

**A5**

| c.c. | s.c. |
|---|---|
| (2) | (2) |

# Symmetry Elements

## SYMMETRY ELEMENTS

SYMMETRY
- POS (Plane of symmetry) ($\sigma$)
- COS (Centre of symmetry or centre of inversion) ($C_i$)
- AOS (Axis of symmetry) ($C_n$)
- AAOS (Alternative axis of symmetry) ($S_n$)

### PLANE OF SYMMETRY (POS) ($\sigma$)

In compound POS may or may not be present.

In a molecule more then one POS may be present.

While imaging POS thickness of molecule is also considered.

Those molecule which are planar, they may cut along their thickness which is known as molecular plane (slice cut).

**Example:**

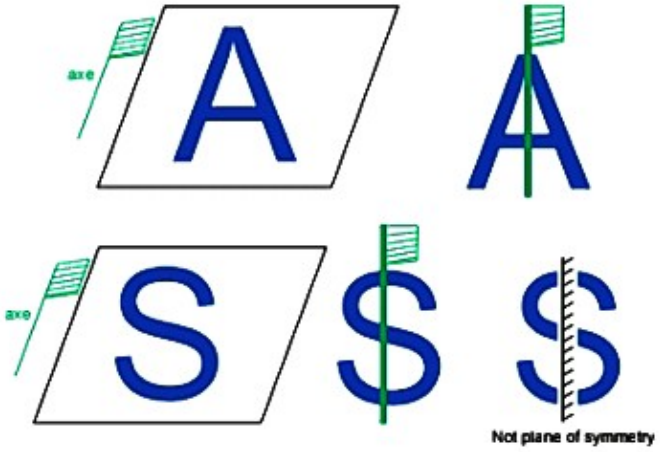
Bread

**Definitions**

- It is an imaginary plane which bisect the molecule in exact two equal half and they are mirror images of each other.



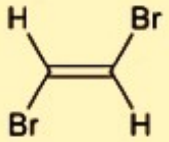
Following alphabetical letters may be used for elementry understanding of P.O.S. :



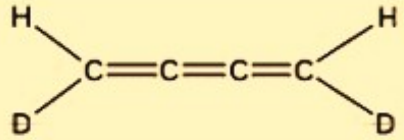
Identify number of following letters which have plane of symmetry.

A B C D E F G
H I J K L M N
O P Q R S T U
V W X Y Z

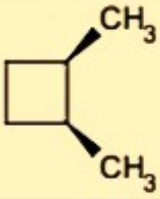
**Identify POS in given molecules :**

**Q1** 


**A1** POS ; Molecular Plane

**Q2** 

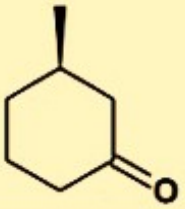

**A2** POS ; V, H, M.P.

**Q3** 

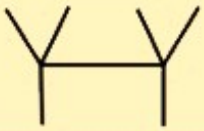

**A3** POS ; H✓ ; V ; M.P. ×

**Q4** 

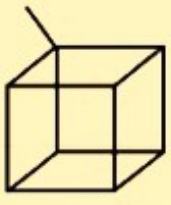
**A4** POS ; H× ; V✓

**Q5** 


**A5** POS ×

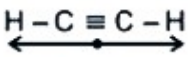
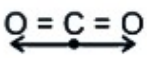
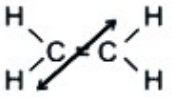
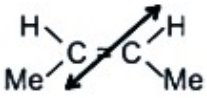
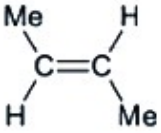
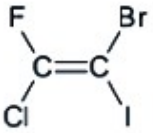
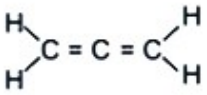
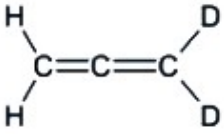
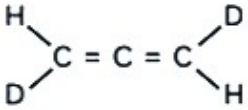
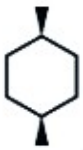
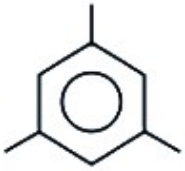
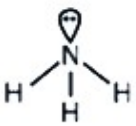
**Q6** 

**A6** POS; Perpendicular plane mid bisect

**Q7** 

**A7** POS; diagonal bisector

## CENTRE OF SYMMETRY (COS)

1.  COS ✓
2.  COS ✓
3.  COS ✓
4.  COS ×
5.  COS ✓
6.  COS ×
7.  COS ✓
8.  COS ×
9.  COS ×
10.  COS ×
11.  COS ✓
12.  COS ✓
13.  COS ×
14.  COS ×
15.  COS ×
16.  COS ✓
17.  COS ×
18.  COS ×

**Definitions**

- A imaginary point in the plane from if we draw two lines in opposite direction meets like atom/group at equal distance, called COS.

**Points to remember**

- COS is absent in eclipsed conformer.
- COS is absent in fischer projection.
- COS may present in staggered conformer.

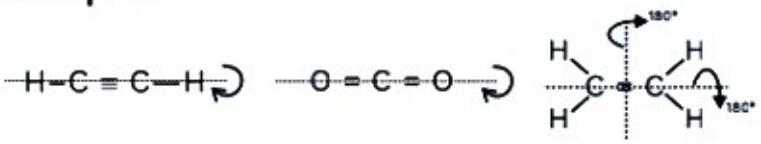
## AXIS OF SYMMETRY (AOS) ($C_n$)

n = 360°

At 360° molecule repeat itself and that axis is called natural axis and it is not considerable.

Linear molecule has infinite AOS.

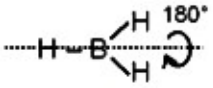
**Example :**



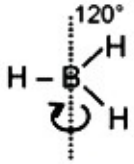
**Definitions**

- If a compound is rotated by 360° by n, about an imaginary axis and if we get in-distinguishable compound (configuration repeat itself) then it is called axis of symmetry.

Axis where value of n is maximum are known as Principle axis or Main axis.



n = 2

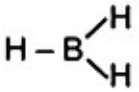


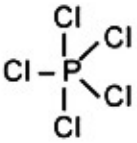
n = 3

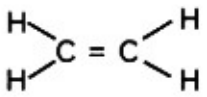
| | | | |
|---|---|---|---|
| If q = 180° | then | n = 2 | $C_2$ |
| If q = 120° | then | n = 3 | $C_3$ |
| If q = 90° | then | n = 4 | $C_4$ |
| If q = 72° | then | n = 5 | $C_5$ |
| If q = 60° | then | n = 6 | $C_6$ |

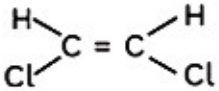
**Q. Find out A.O.S. in following molecules :**

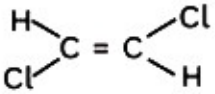
1. O = C = O

2. 

3. 

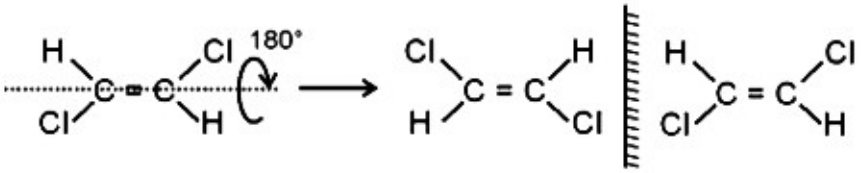
4. 

5. 

6. 

## ALTERNATE AXIS OF SYMMETRY (AAOS) ($C_N$)

**Example :**



**Example:**

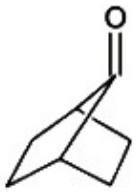
**Q.** **Comment about POS, COS and AOS in following molecules :**

**1.** $CH_4$

**Ans.** POS (s) → ✓
$C_2$ → ✓
$C_3$ → ✓
$C_i$ → ×

**2.** 

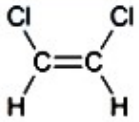
**Ans.** POS (s) → ✓
$C_2$ → ✓
$C_3$ → ✓
$C_i$ → ✓

**3.** 

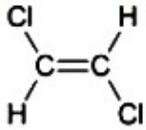

**Ans.** POS (s) → ✓
$C_2$ → ✓
$C_3$ → ×
$C_i$ → ×

**4.** 

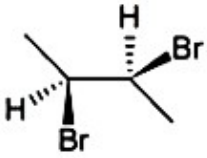
**Ans.** POS (s) → ✓
$C_2$ → ✓
$C_3$ → ×
$C_i$ → ×

**5.** 

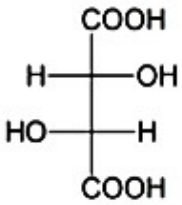

**Ans.** POS (s) → ✓
$C_2$ → ✓
$C_3$ → ×
$C_i$ → ×

**6.** 


**Ans.** POS (s) → ✓
$C_2$ → ✓
$C_3$ → ×
$C_i$ → ✓

**7.** 


**Ans.** POS (s) → ×
$C_2$ → ✓
$C_3$ → ×
$C_i$ → ×

**8.** 

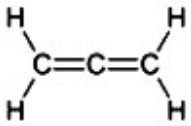

**Ans.** POS (s) → ×
$C_2$ → ✓
$C_3$ → ×
$C_i$ → ×

**DPP-23**

**Q.** **Answer about following molecules in below pattern :**

Chiral Carbon → _ _ _ (✓/×)
POS → _ _ _ (✓/×)
COS → _ _ _ (✓/×)
Geometrical isomerism → _ _ _ (✓/×)

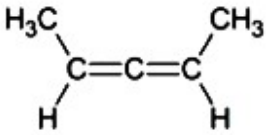
**1.**



**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ×

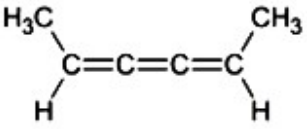
**2.**



**Ans.**

Chiral Carbon → ×
POS → ×
COS → ×
Geometrical isomerism → ×

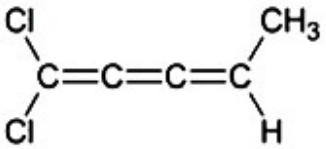
**3.**



**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ✓

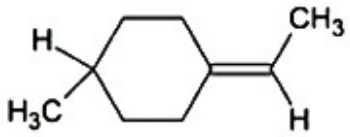
**4.**



**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ×

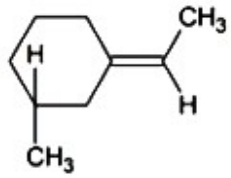
**5.**



**Ans.**

Chiral Carbon → ×
POS → ×
COS → ×
Geometrical isomerism → ×

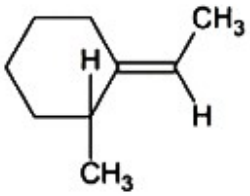
**6.**



**Ans.**

Chiral Carbon → ✓
POS → ×
COS → ×
Geometrical isomerism → ✓

**7.**



**Ans.**

Chiral Carbon → ✓
POS → ×
COS → ×
Geometrical isomerism → ✓

**8.** 

**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ×

**9.** 


**Ans.**

Chiral Carbon → ×
POS → ×
COS → ×
Geometrical isomerism → ×

**10.** 

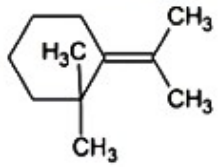

**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ×

**11.** 

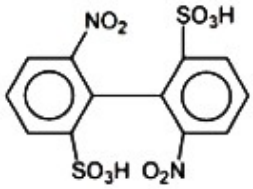

**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ✓

**12.** 

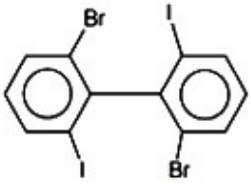

**Ans.**

Chiral Carbon → ×
POS → ✓
COS → ×
Geometrical isomerism → ×

**13.** 


**Ans.**

Chiral Carbon → ×
POS → ×
COS → ×
Geometrical isomerism → ×

**14.** 


**Ans.**

Chiral Carbon → ×
POS → ×
COS → ×
Geometrical isomerism → ×

# R–S system (Absolute configuration)

**R–S SYSTEM (ABSOLUTE CONFIGURATION)**

R → Rectus (Right)

S → Sinister (Left)

**R-S nomenclature is assigned as follow :**

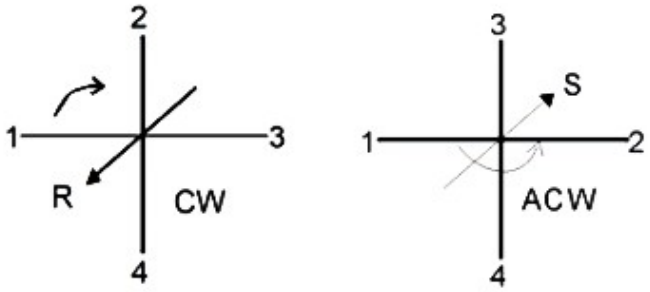
**Step-I :-** By the set of sequence rule, we give the priority order of atom or group connected through the chiral carbon.

**Step-II :-** If atom/group of minimum priority present on the vertical line, then

Movement of eyes in clockwise direction = R

Movement of eyes in anticlockwise = S

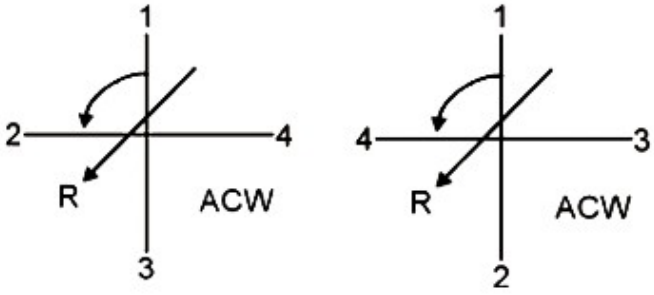
Movement of eyes taken from 1 → 2 → 3 through low molecular weight group (if needed)



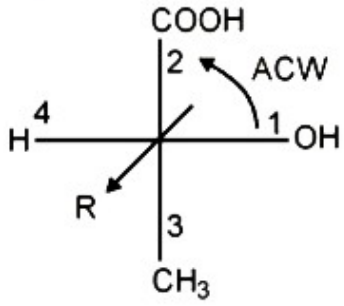

**Step III :-** If minimum proirity group present on the horizontal line, then

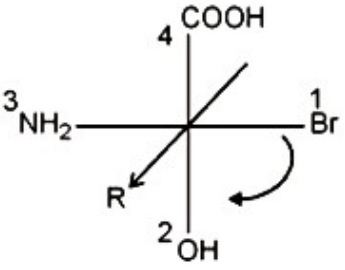
clockwise rotation ⇒ S

anticlockwise rotation ⇒ R

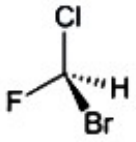


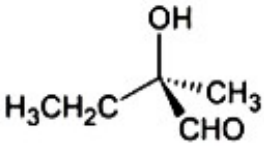

**Example:**

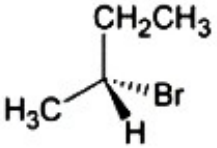
1. 

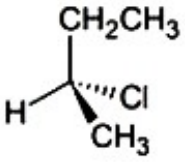

2. 

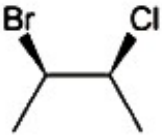
**Q1** **Indicate whether each of the following structure has the R-configuration or the S-configuration :**

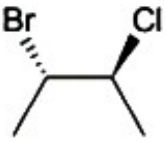
1.  → S

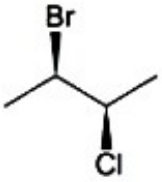
2.  → R

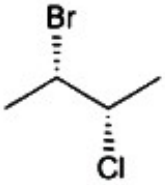
3.  → R

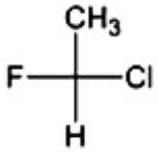
4.  → S

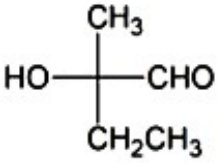
5.  → 2R–3S

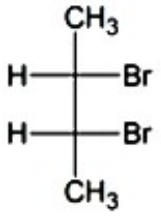
6.  → 2S–3S

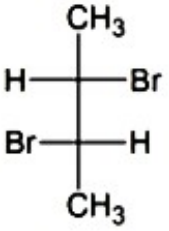
7.  → 2R–3R

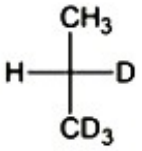
8.  → 2S–3S

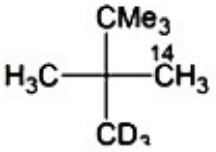
9.  → R

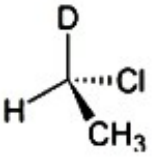
10.  → R

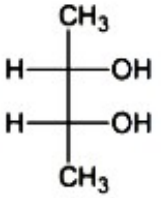
11.  → 3S–3R

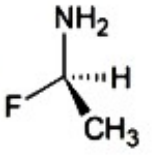
12.  → 2S–3S

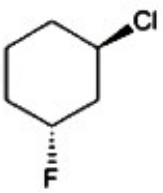
13.  → S

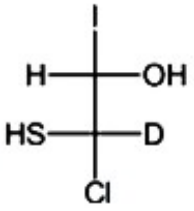
14.  → R

15.  → R

16.  → 2S–3R

17.  → R

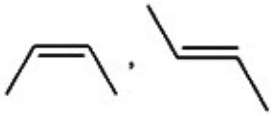
18.  → 1R–3R

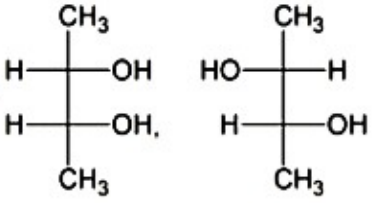
19.  → 1S–2S

# Different Relationships of Stereoisomers

**DIASTEREOMER**

**Example :**

cis-trans 

Erythro-threo 

**Definitions**

- Those stereoisomers which are not mirror image of each other.

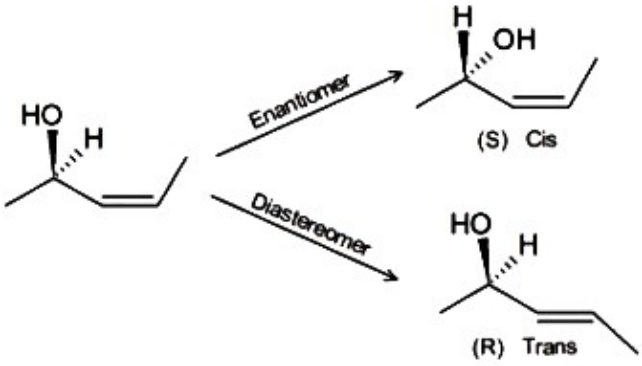
Two chiral centre + one interchange

RR
- SS ⟶ Enantiomer
- RS ⟶ Diastereomer
- SR — Diastereomer

Three chiral centre + two interchange

RR
- SSS ⟶ Enantiomer
- RRS ⟶ Diastereomer
- RSR ⟶ Diastereomer
- RSS ⟶ Diastereomer

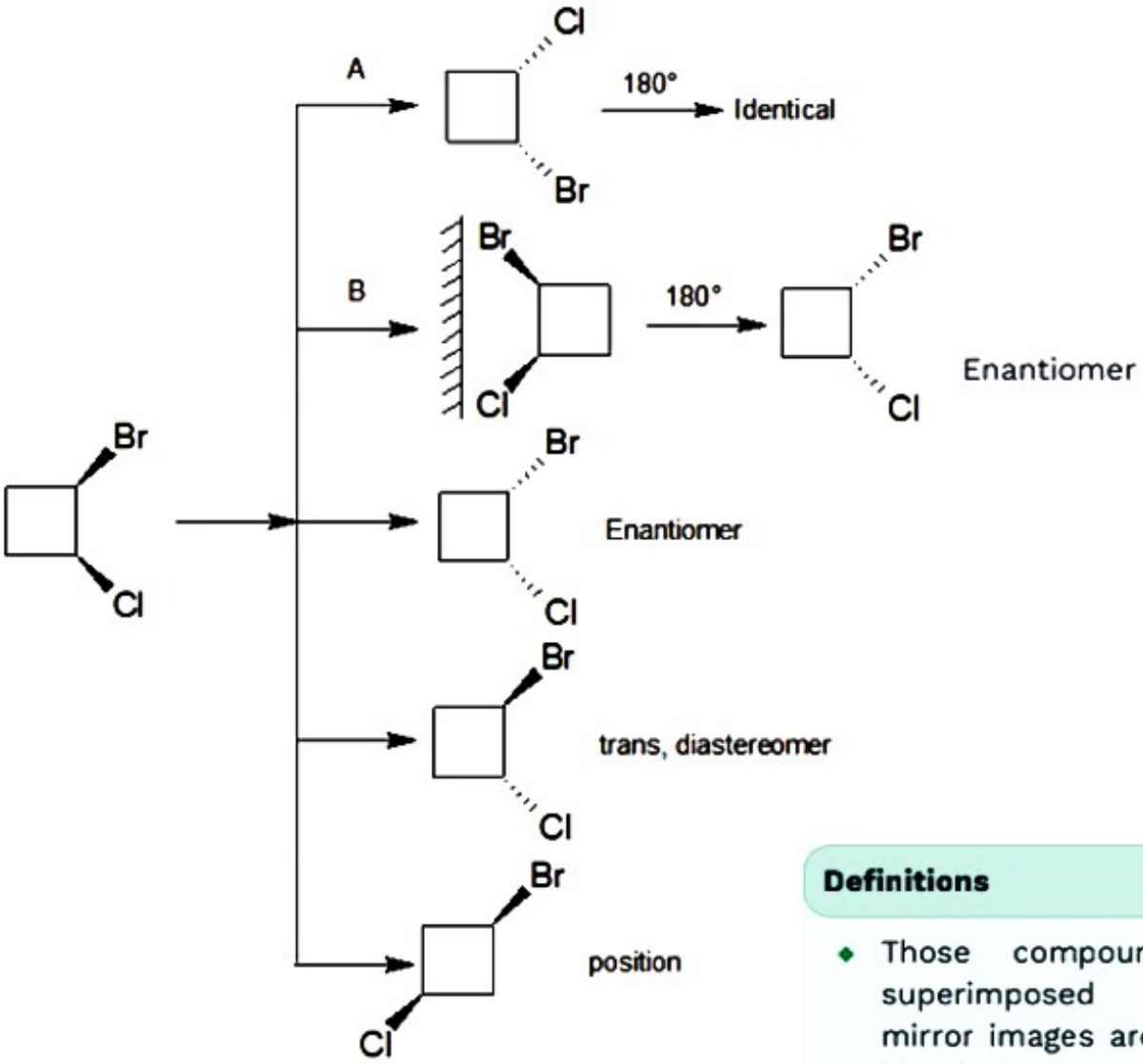
**Example:**




**Tips and Trick**

- Diastereomer having different physical properties melting, boiling point, solubility.
- Diastereomer are having different chemical properties.
- If only one chiral centre then diastereomer is not possible but if there is chiral centre with π bond then diasteromer possible.

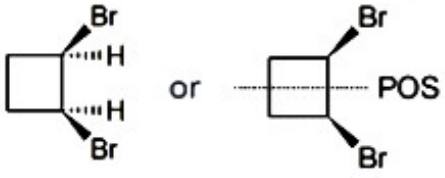
**Example:**




**IDENTICAL**

(i) COS / centre of symmetry / $C_i$
(ii) POS / plane of symmetry / s
(iii) AOS / axis of symmetry / $C_n$
(iv) AAOS/ Alternate axis of symmetry / $S_n$

Meso = two or more C.C. + POS / COS / AAOS




**Note:** Meso compound are optically inactive but consider in optical isomer because it rotates PPL light but the net result is zero.

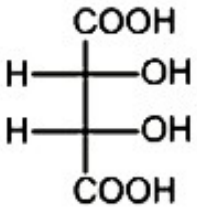
**Definitions**

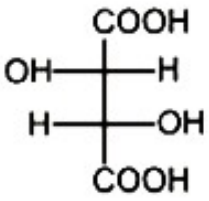
- Those compounds which superimposed on their mirror images are known as identical.

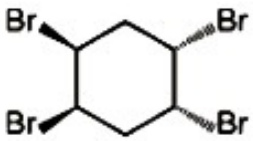
**Points to remember**

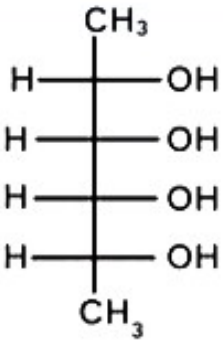
- Those compounds which are having chiral centre even though optically inactive.
- Meso compound are optically inactive because half compound rotates PPL in clockwise direction and remaining half part in anticlockwise direction.

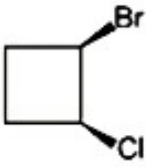
## Q1 Identify meso :

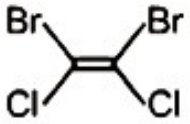
1. 

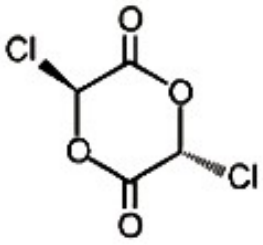

2. 


3. 


4. 


5. 


6. 

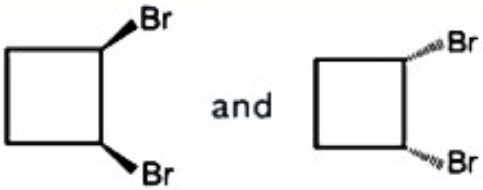

7. 

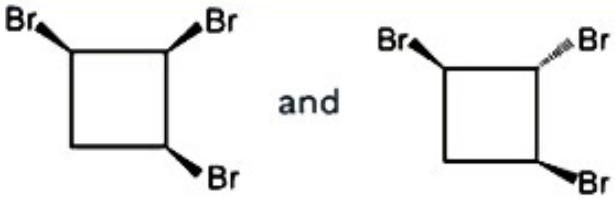

**Solutions :**

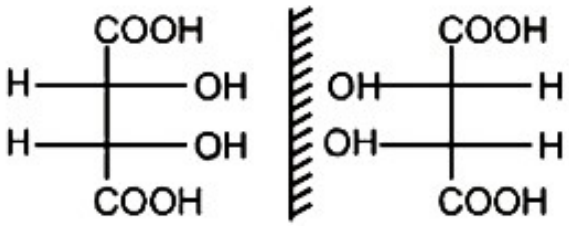
1. Meso
2. Optically active
3. Meso
4. Meso
5. Optically active
6. Geometrical Isomer, Optically inactive
7. Optically active

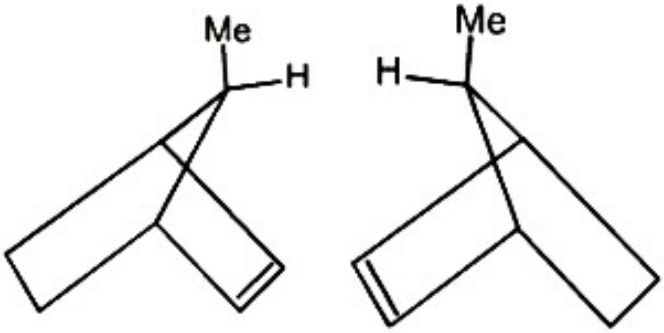
**Note :** Meso compounds are identical or diastereoisomer but not enantiomer.

## Q2 Identify following meso or diastereoisomer

1. 


2. 


3. 


4. 

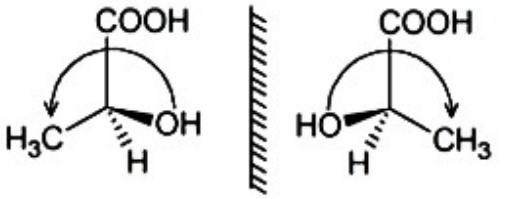

**Answer :**

1. Identical
2. diastereoisomers
3. Identical
4. Identical

**ENANTIOMER**

Superimpose A | A

Non super impose B | B

Anti 

Clockwise

**Definitions**

- Those stereoisomers which are mirror image of each other but do not superimpose on each other are known as enantiomer.

Enantiomers are having same physical properties like melting point, boiling point, solubility except rotation towards plane polarised light.

Enantiomers are having same chemical properties except reaction with enzyme.

**Tips and Trick**

- Equimolar mixture of enantiomer is known as

  Racemic mixture = 50% d + 50% $\ell$

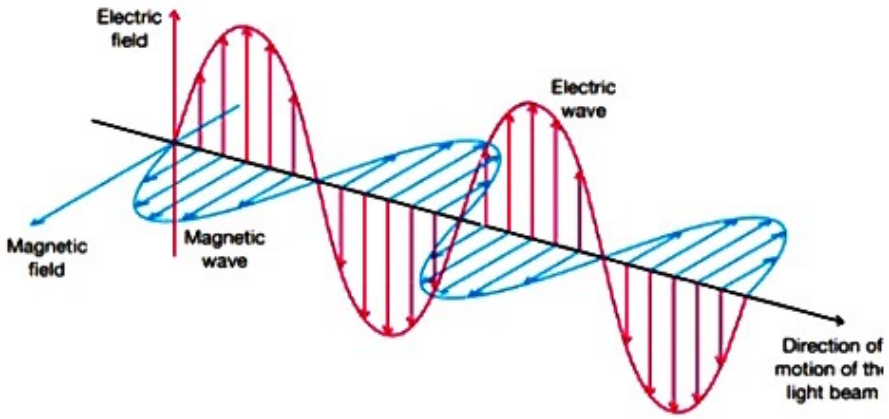
  Optically inactive

# Chiral-Achiral Molecule

**POLARIMETER**

**CHIRAL-ACHIRAL MOLECULE**

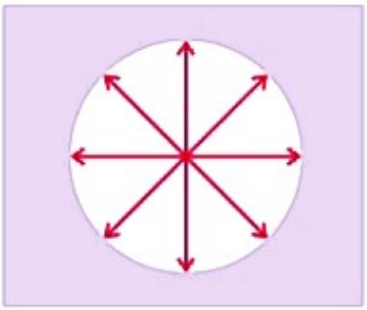
**ELECTROMAGNETIC WAVE OF LIGHT :**



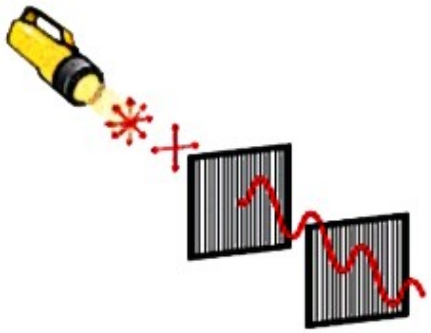

**OPTICAL ACTIVITY**

**Introduction**

- Substances which can rotate the plane-polarised light are called optically active substances and the property of a substance to rotate the plane-polarised light is called optical activity.



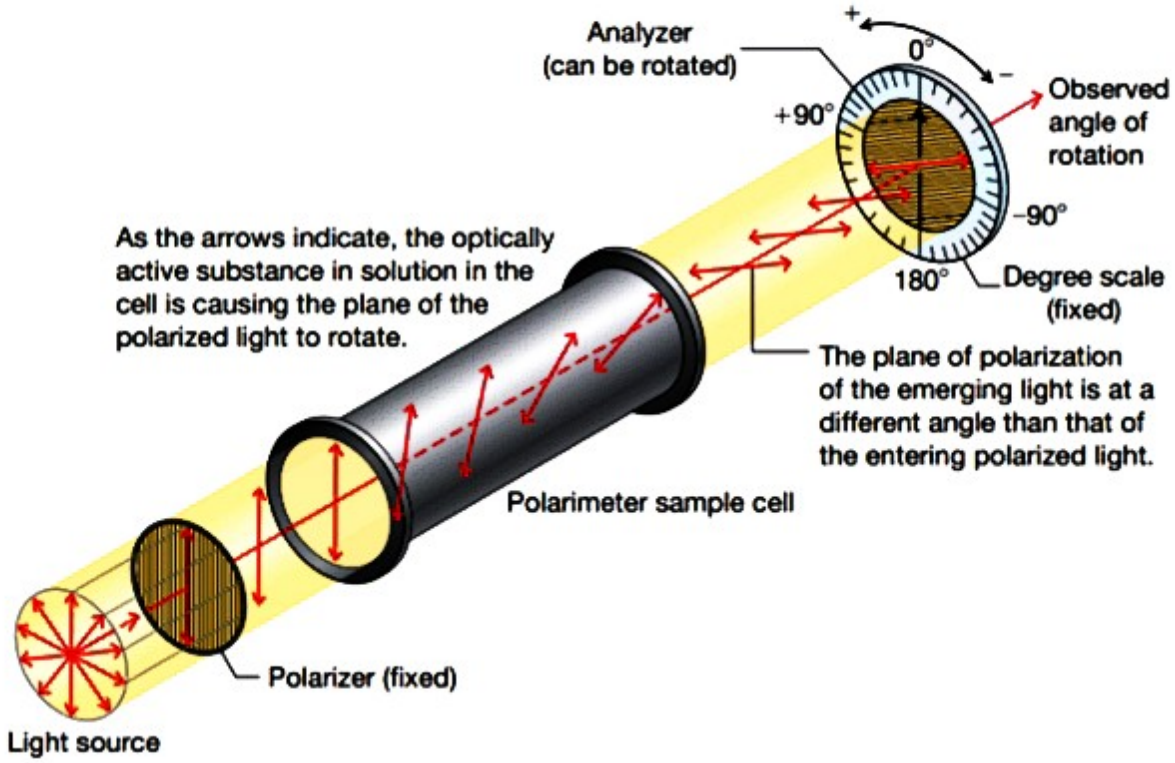
Normal light



Basic understanding of PPL using slit

**Definitions**

- The instrument used for measuring the optical rotation called a polarimeter. It consists of two Nicol prisms, one called the polariser (near the light source) and the other called the analyser, a glass tube containing the solution of an optically active compound is placed.

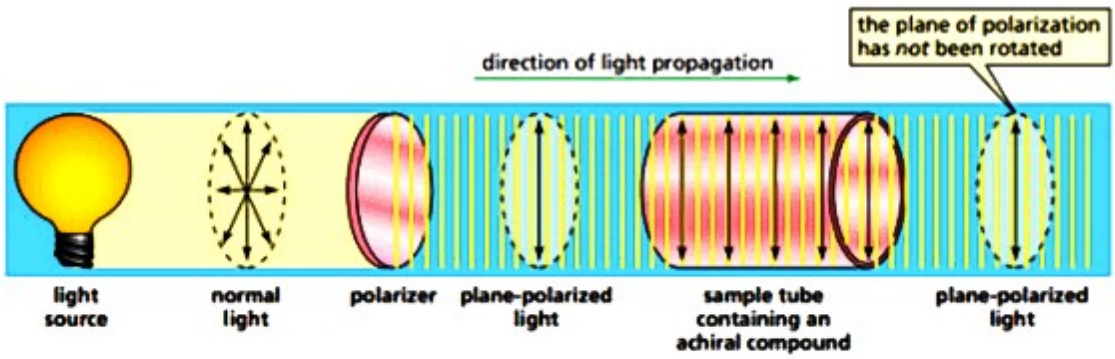


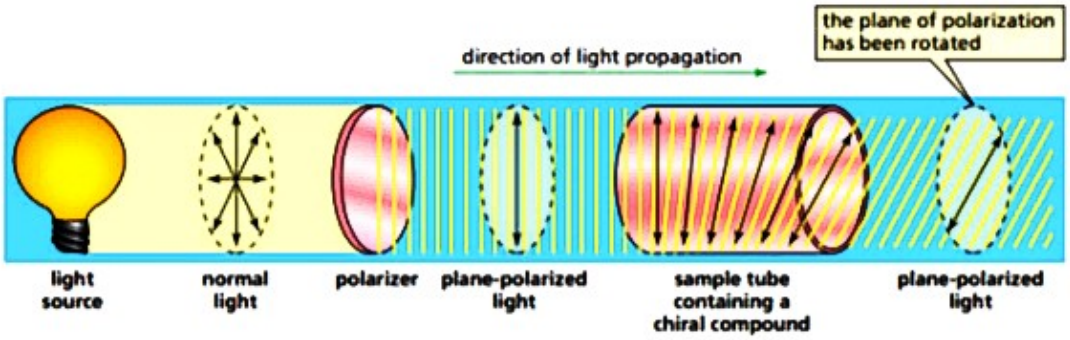

## OPTICAL ISOMERISM

**Let's understand**

A substance which rotates the plane-polarised light towards right (in the clockwise direction) is called dextrorotatory (Latin : dexter = right) and is usually represented by the prefix d or (+). On the other hand, a substance which rotates the plane-polarised light towards the left (in the anticlockwise direction) is called laevorotatory (Latin : leaves = left) and is usually represented by the prefix 'l' or (–).

Behaviour of Optical Active and Inactive Molecules towards PPL






**Chirality of molecules and objects :**

An inspection of the molecular structures of optically active compounds indicates that all these compounds have Chiral* (or dissymmetric) molecules.

A molecule (or an object) is said to be chiral or dissymmetric, if it is not superimposable on its mirror image, and the property of non-superimposability is called chirality.

On the other hand, a molecule (or an object) which is superimposable on its mirror image is called achiral (non-dissymmetric or symmetric).

**Specific rotation :**

$[a]_D^t = \frac{\theta}{\ell \times C}$, where $\theta$ is the optical rotation, $\ell$ is the length of the tube in decimetre, C is the concentration of the solution in g $ml^{-1}$, D is the wavelength of sodium light, and t is the temperature.

**Racemic forms and Enantiomeric Excess :**

A sample of an optically active substance that consists of a single enantiomer is said to be enantiomerically pure or to have an enantiomeric excess of 100%.

An enantiomerically pure sample of (S)-(+)-2-butanol shows a specific rotation of +13.52°. On the other hand, a sample of (S)-(+)-2-butanol that contains less than an equimolar amount of (R)-(−)-2-butanol will show a specific rotation that is less than +13.52° but greater than 0°. Such a sample is said to have an enantiomeric excess less than 100%. The enantiomeric excess (ee) is defined as follows :

% Enatiomeric excess

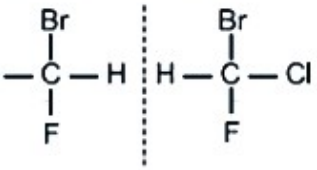
$$= \frac{\text{moles of one enantiomer} - \text{moles of other enantiomer}}{\text{total moles of both enantiomers}} \times 100$$

$$\% \text{ Enantiomeric excess} = \frac{\text{observed specific rotation}}{\text{specific rotation of the enantiomer}} \times 100$$

**Q.** A mixture of the 2-butanol enantiomers showed a specific rotation of +6.76°. We would then say that the enantiomeric excess of the (S)-(+)-2-butanol is 50%.

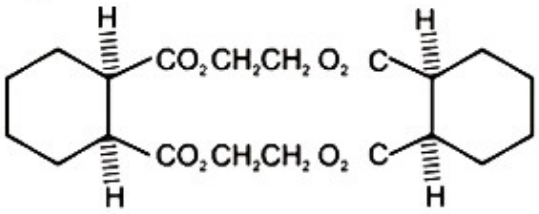
**Sol** Enantiomeric excess = $\frac{+6.76°}{+13.52°} \times 100 = 50\%$

**Conditions for optical activity :**

For example : 

- It must have non-superimposable mirror image.
- It should not contain any element of symmetry, i.e.,
  - **i)** Plane of symmetry or (mirror plane or s-plane).
  - **ii)** Centre of symmetry or centre of inversion.
  - **iii)** Alternating axis of symmetry.

**Example:**

**1.** 

Optically active → ×
Chiral molecule → ×
Achiral molecule → ✓
Optically inactive → ✓

**Reason :**

P.O.S ✓
C.O.S ✓

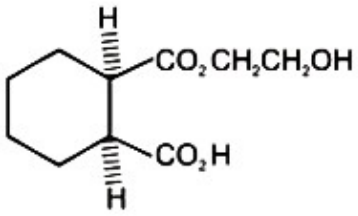
**Introduction**

- It must have an assymmetric C atom or the chiral C atom. C atom which is attached to four different atoms or groups is called an asymmetric C atom or the chiral centre or stereogenic or stereocentre.

**Points to remember**

- Chiral, optically active then POS as well as COS should be absent.
- Achiral, optically inactive then POS and COS both absent.
- Compound to be achiral it should have POS or COS or both.
- There is no relationship between AOS and optically activity of molecule.

2.

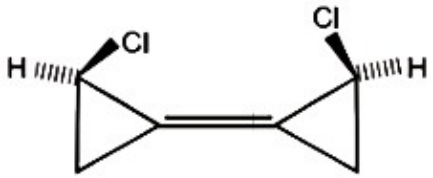



Optically active → ✓
Chiral molecule → ✓
Achiral molecule → ×
Optically inactive → ×

**Reason :**

P.O.S ×
C.O.S ×

3.

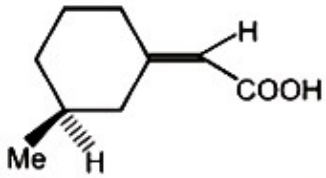



Optically active → ×
Chiral molecule → ×
Achiral molecule → ✓
Optically inactive → ✓

**Reason :**

P.O.S ✓

4.

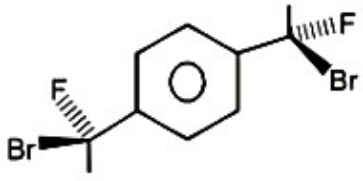



Optically active → ✓
Chiral molecule → ✓
Achiral molecule → ×
Optically inactive → ×

**Reason :**

P.O.S ×
C.O.S ×

5.

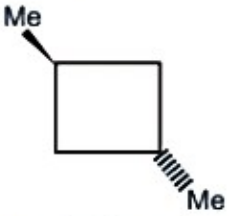



Optically active → ✓
Chiral molecule → ✓
Achiral molecule → ×
Optically inactive → ×

**Reason :**

P.O.S ×
C.O.S ×

6.

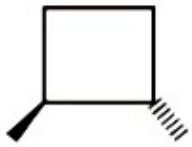



Optically active → ×
Chiral molecule → ×
Achiral molecule → ✓
Optically inactive → ✓

**Reason :**

P.O.S ✓
C.O.S ✓

7.

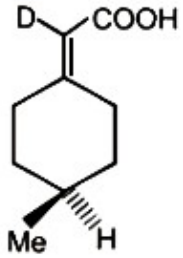


Optically active → ✓
Chiral molecule → ✓
Achiral molecule → ×
Optically inactive → ×

**Reason :**

P.O.S ×
C.O.S ×

8.

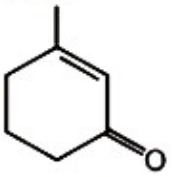



Optically active → ✓
Chiral molecule → ✓
Achiral molecule → ×
Optically inactive → ×

**Reason :**

P.O.S ×

C.O.S ×

9. 

Optically active → ✓

Chiral molecule → ✓

Achiral molecule → ×

Optically inactive → ×

**Reason :**

P.O.S ✓

**10. Hockey shaped molecule**

Optically active → ×

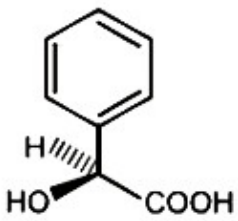
Chiral molecule → ×

Achiral molecule → ✓

Optically inactive → ✓

**Reason :**

P.O.S ✓

11. 

Optically active → ✓

Chiral molecule → ✓

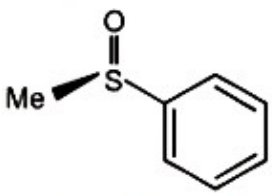
Achiral molecule → ×

Optically inactive → ×

**Reason :**

P.O.S ×

C.O.S ×

12. 

Optically active → ×

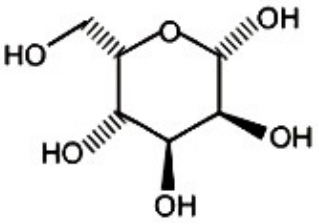
Chiral molecule → ×

Achiral molecule → ✓

Optically inactive → ✓

**Reason :**

P.O.S ✓

13. 

Optically active → ✓

Chiral molecule → ✓

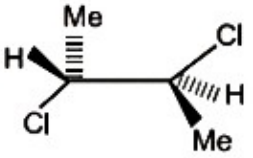
Achiral molecule → ×

Optically inactive → ×

**Reason :**

P.O.S ×

C.O.S ×

14. 

Optically active → ×

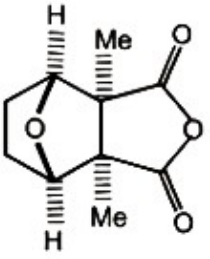
Chiral molecule → ×

Achiral molecule → ✓

Optically inactive → ✓

**Reason :**

C.O.S ✓

15. 

Optically active → O

Chiral molecule → O

Achiral molecule → ✓

Optically inactive → ✓

**Reason :**

P.O.S ✓

# D/L, Epimer, Anomer / Erythro-Threo / Amine flipping

**D/L NOMENCLATURE :**

**Rule :**

**(i)** This nomenclature is only applicable for fischer projection of carbohydrate and amino acid.

**(ii)** Carbohydrate means optically active polyhydroxy carbonyl compound.

**(iii)** Amino acid is the monomer of protein.

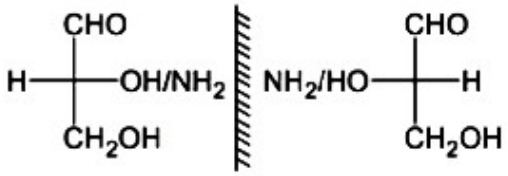
**(iv)** For D-L nomenclature, fischer must be correct.

**Condition :**

(a) Carbon containing main functional group at top

(b) Maximum carbon on vertical line.

(c) For one chiral centre if OH/$NH_2$ group on right side then D and if on left side then L.

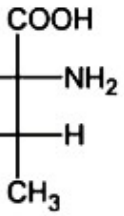



**D** **L**

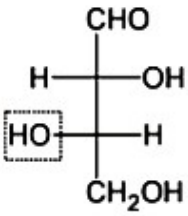
(d) If two or more chiral centre are present then in amino acid consider first chiral centre and in carbohydrate consider last chiral centre for D, L configuration.

**Q. Identify the D / L configuration**

**1.**

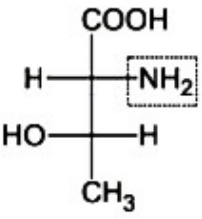



**Sol.**



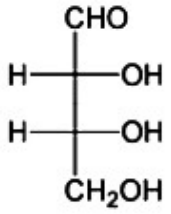

L-Carbohydrate

**Sol.**

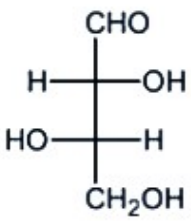



Right; D-Amino

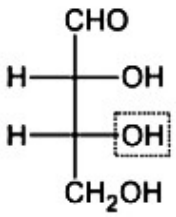
**3.**




**2.**

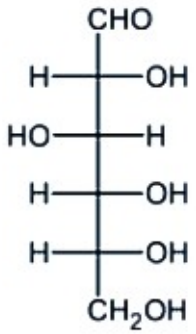



**Sol.**

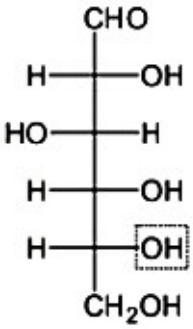



D-Carbohydrate

**4.**

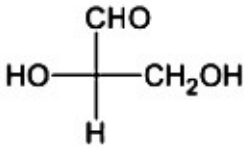



**Sol.**

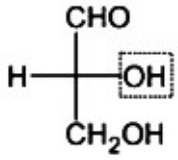



D-Glucose

**5.**

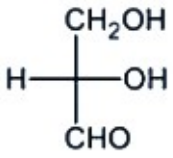



**Sol.**

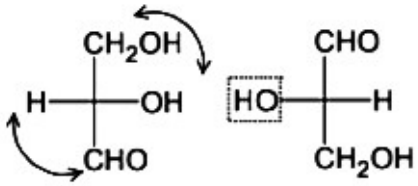



D-Carbohydrate

**6.**




**Sol.**




L-Carbohydrate

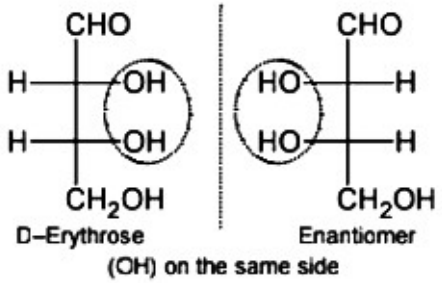
## ERYTHRO AND THREO DESIGNATION

The stereoisomers in which the same groups are present on the same side of the Fischer projection are called erythro isomers while the stereoisomers in which these groups are present on the opposite sides are known as threo isomers.

**Definitions**

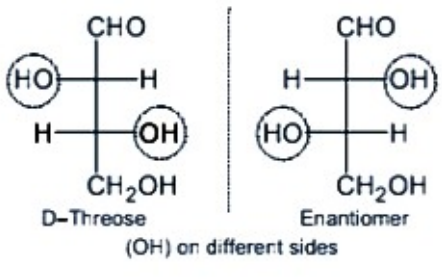
- Compound with two adjacent chiral (stereogenic) C atoms are sometimes designated by prefix 'erythro or threo'.

**Example-1 :**



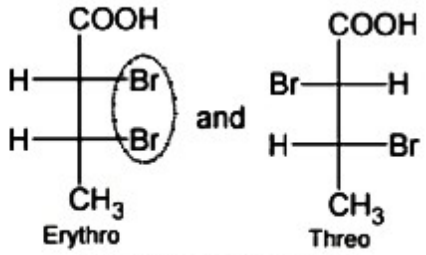

(OH) on the same side

**Example-2 :**




(OH) on different sides

**Example-3 :**



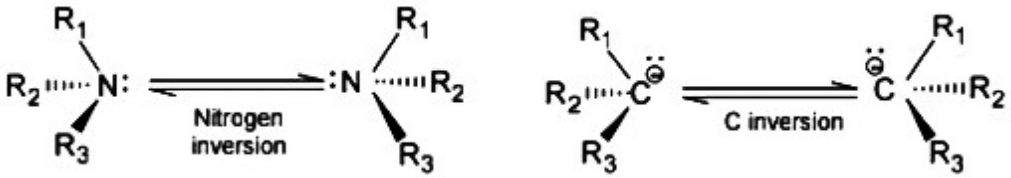
2,3-Dibromo butanoic acid

**OPTICAL ACTIVITY OF TERT-AMINES AND TERT-CARBANION OF THE TYPE ($R_1R_2R_3N$:) AND ($R_1R_2R_3:C^-$)**

The energy required for this inversion is very low at room temperature, since this inversion does not involve bond breaking and bond formation. Thus racemisation occurs and enantiomers cannot be isolated.

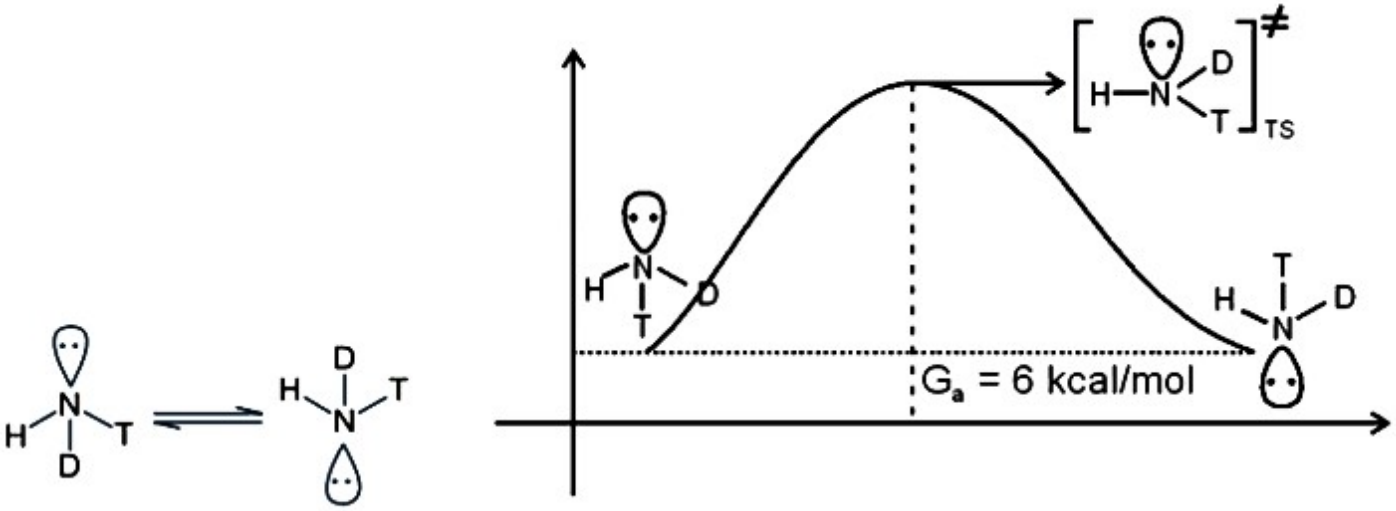
**Let's understand**

Such an amine and carbanion with a lone pair of electrons (being considered a fourth different group) are chiral because of their pyramidal geometry, but they are optically inactive. A rapid 'umbrella' type inversion converts either of the enantiomers to a racemic mixture.

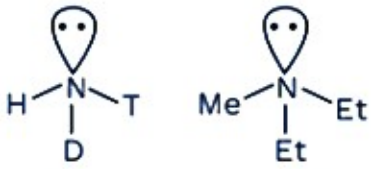



But amine salts are optically active. Nitrogen inversion in not possible, e.g., $[R_1R_2R_3N^+H]Cl^-$.

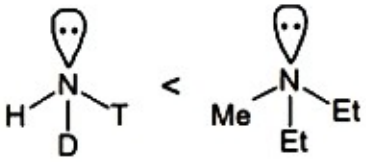
**AMINE FLIPPING (UMBRELLA INVERSION / AMINE INVERSION)**



**Q1** **Why compound (B) has rapid inversion tendency compare to compound (A)**




(A) (B)

**Sol** 

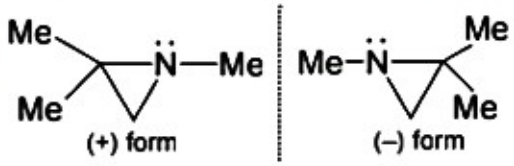

Bulky group

(A) (B)

Bulky valencies at $sp^3$ hybridised nitrogen atom tends more flipping.

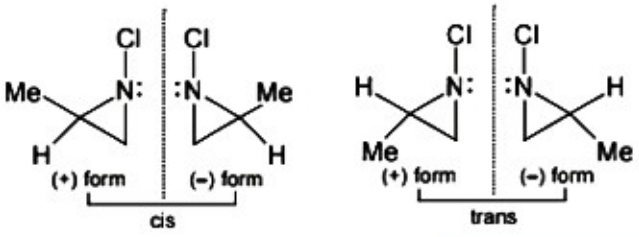
$\Delta G$ (A) > (B)

**SOME MORE EXAMPLES OF OPTICALLY ACTIVE N-CONTAINING COMPOUNDS :**

**(i)** 


1,2,2-Trimethyl aziridine

**(ii)** 1-Chloro-2-methyl aziridine shows four stereoisomers.




**RACEMIC MIXTURE**

The optical activity of a racemic mixture is due to external compensation, i.e., the rotation caused by molecules of one enantiomer is exactly cancelled by an equal and opposite rotation caused by the same number of molecules of the other enantiomer.

**Introduction**

- When equal amounts of two enantiomers are mixed together, it gives an optically inactive form called racemic mixture or modification.
- It is denoted by using the prefix ± or dl before the same of compound.

**Points to remember**

- A racemic mixture can also be obtained chemically. The process by which one enantiomer (+ or –) of an optically active compound is converted into a racemic mixture is called racemisation. Racemisation can be carried out by heat, light or chemical reagents.

**Example :**

Lactic acid and Tartaric acid

**Resolution of Racemic Mixture :**

- Enantiomers have identical physical and chemical properties except towards optically active reagents, they cannot be separated by the usual techniques such as fractional crystallisation, fractional distillation, chromatography, etc. Therefore, special methods are used to achieve their separation. Some of these are listed below :
  - **i)** Mechanical method
  - **ii)** Biological method
  - **iii)** Chemical methods (by making their diastereomers)
  - **iv)** Chromatographic separation by using special adsorbents.

**Diastereomeric method**

These can be separated from one another by conventional methods of separation of compounds. The separated diastereoisomer is then broken down to give pure enantiomers.

Chemically we can seperate enantiomer via diastereomerism

**i)** d + l + d or l ⟶ d – d + l – l
(Racemic mixture) (Diastereomer)

(+) + (–) + (+) or (–) ⟶ (+, +) + (+, –)
(Diastereomer)

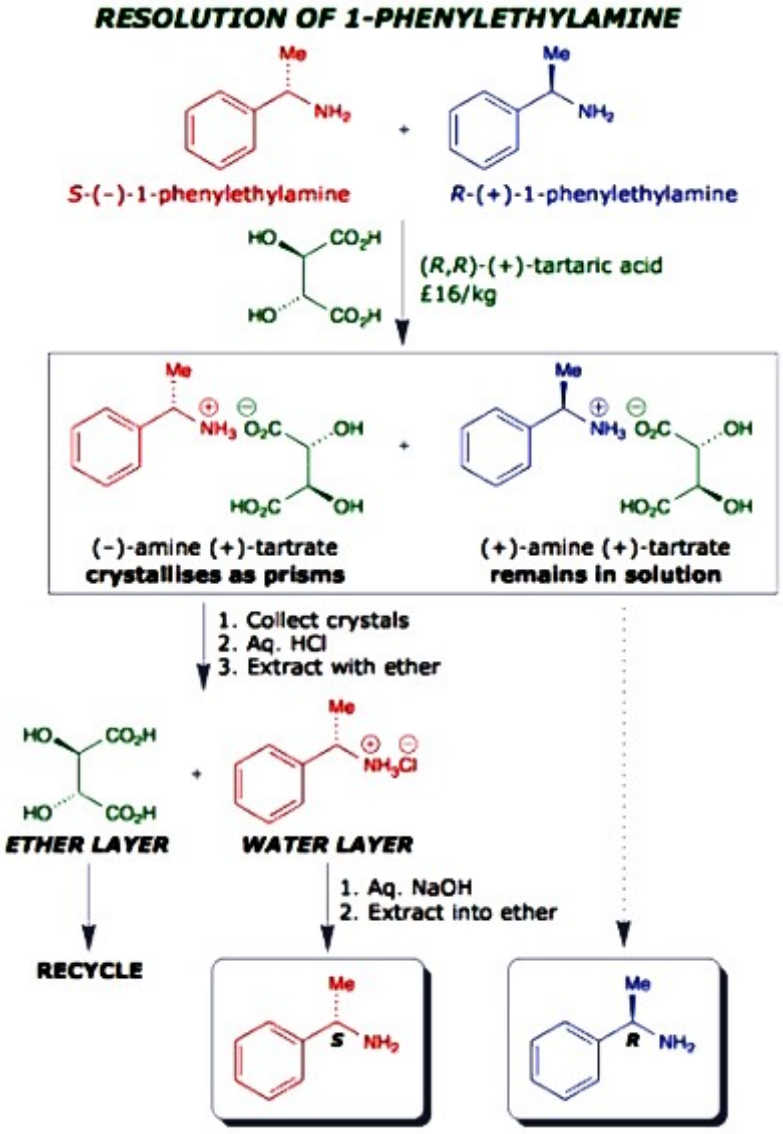
**Definitions**

- The process of separation of racemic mixture into its (+) and (–) enantiomers is called resolution.

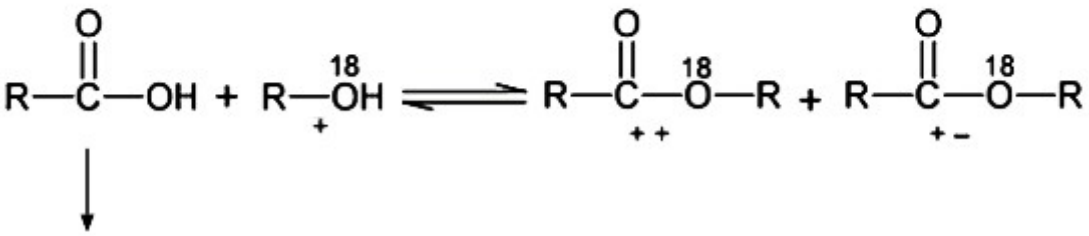
**Introduction**

- One of the most common method is to allow a racemic mixture to react with an enantiomer of some other compound. This changes a racemic form into a mixture of diastereomers which have different solubilities and melting and boiling points.

**Example-1 :**




**Example-2 :**

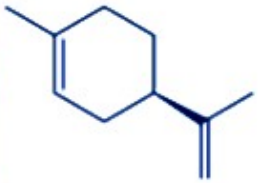


(Racemic Mixture, + −)

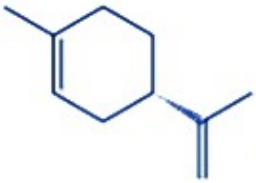
**The importance of stereochemistry :**

Chiral molecules can show their handiness in many ways, including the way they affects human being. One enantiomeric form of a compound called limonene is primarily responsible for the odor of oranges and the other enantiomer for the odor of lemons.





**(+)-Limonene (the enantiomer of limonene found in oranges)**



**(–)-Limonene (the enantiomer of limonene found in lemons)**



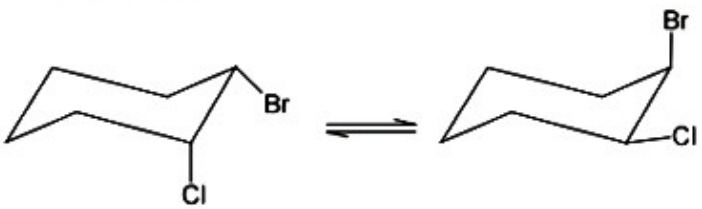
- All naturally occurring sugars, including the sugars that occur in DNA, are of D-configuration. the yeast enzymes can specifically ferment D-glucose and not its L-enantiomer.
- Stereochemistry also plays an important role in deciding the physiological properties of compounds.

  (–)-Nicotine is much more toxic than (+)-Nicotine, (+)-adrenaline is very active in construction of blood vessels than (–)-adrenaline.
- Chirality is crucial for the effect of drugs as well. In majority of cases, only one enantiomer is found to have the desired effect while the other isomer may be totally inactive or has an opposite effect.

**Example :**

It is the (S)-enantiomer of ibuprofen that has the pain-relieving action. (–)-Thyroxine, an amino acid of thyroid gland, speeds up metabolic processes and causes nervousness and loss of weight. Its enantiomer, (+)-thyroxine, has none of these effects but is used to lower the cholesterol levels.

**Ring Flipping**




Non resolvable

# Calculation of Optical Isomer (O.I.)

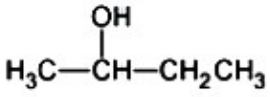
**CASE-I : FOR UNSYMMETRICAL MOLECULES**

For calculation of optical isomers, in unsymmetrical molecule always apply

Total Number of Optical Isomers = $2^n$

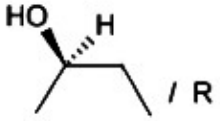
where, n = Number of Chiral Centre / Number of even allene system / spiro / biphenyl

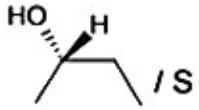
**Q1** **Calculate number of optical isomer (if possible) :**

**1.** 

**Sol.** n = 1 chiral centre

$2^1 = 2$

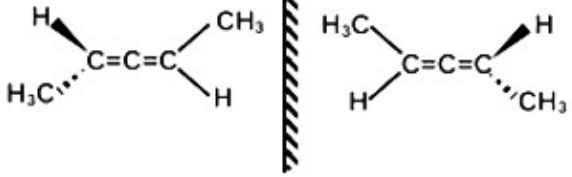


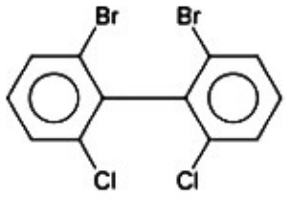




**2.** $CH_3 - CH = C = CH - CH_3$

**Sol.** n = 1 chiral centre

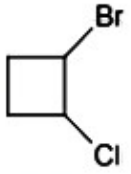
$2^1 = 2$




**3.** 

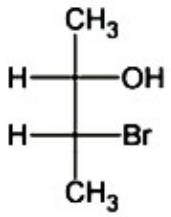

**Sol.** n = 1 chiral centre

$2^1 = 2$

**4.** 

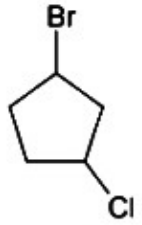

**Sol.** n = 2 chiral centre

$2^2 = 4$

**5.** 

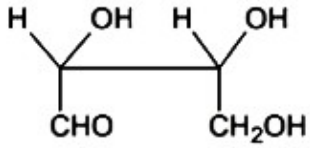

**Sol.** n = 2 chiral centre

$2^2 = 4$

**6.** 


**Sol.** n = 2 chiral centre

$2^2 = 4$

**7.** 

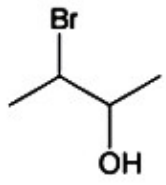

**Sol.** n = 2 chiral centre

$2^2 = 4$

**8.** 


**Sol.** n = 2 chiral centre

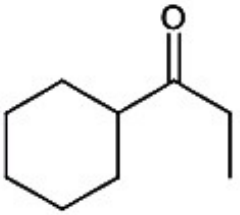
$2^2 = 4$

**9.** 

**Sol.** n = 2 chiral centre

$2^2 = 4$

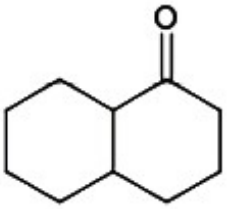
**10.**




**Sol.** n = 0 chiral centre

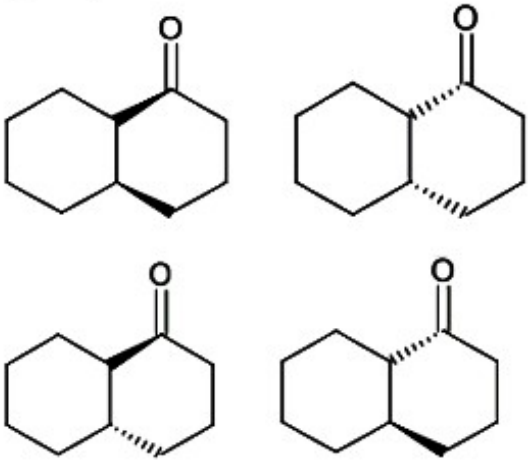
No optical isomerism possible

**11.**

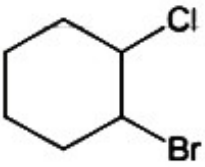



**Sol.** n = 2 chiral centre

$2^2 = 4$

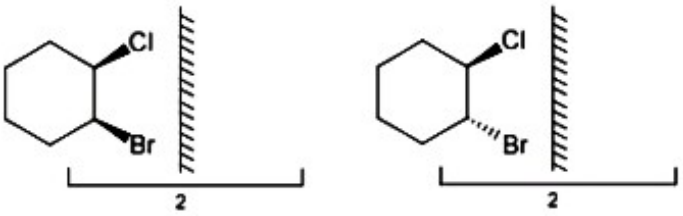



**12.**



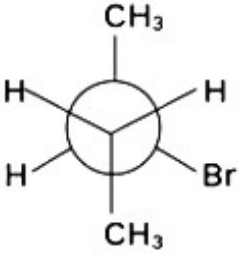

**Sol.** n = 2 chiral centre

$2^2 = 4$



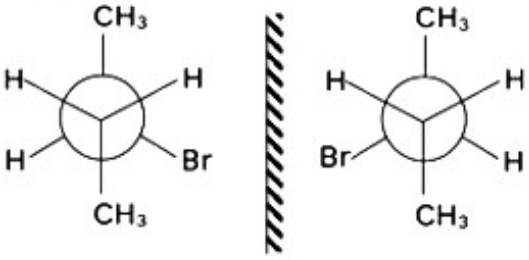

If optical isomer and geometrical isomer same centre then consider optical isomer here geometrical isomer included.

**13.**



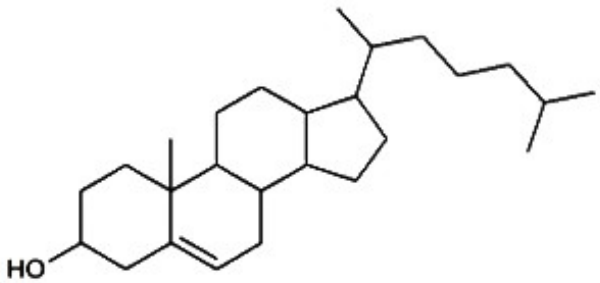

**Sol.** n = 1 chiral centre

$2^1 = 2$




## Q2 Number of possible optical isomers for following molecules :

**1.**




**Sol.** n = 8 chiral centre

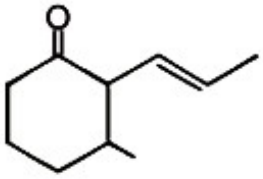
Optical isomer = 256

**2.** 2,4-dichloropentane

**Sol.** n = 2 chiral centre

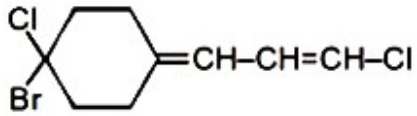
Optical isomer = 4

**3.**



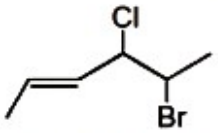

**Sol.** n = 2 chiral centre

Optical isomer = 4

**4.** 

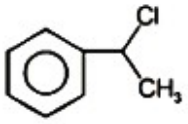

**Sol.** n = 1, spirane system

Optical isomer = 2

**5.** 


**Sol.** n = 2 chiral centre

Optical isomer = 4

**6.** 


**Sol.** n = 1 chiral centre

Optical isomer = 2

**CASE-II :**

Calculation of optical isomers in Symmetrical molecule :

Condition : POS or / and COS present

n = Even

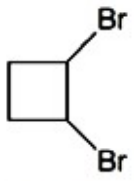
meso + optically active = optical isomer

$2^{\left(\frac{n}{2}-1\right)} + 2^n - 1$ = Total Optical Isomers

n = Odd

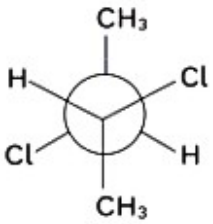
We suggest to draw optical isomer / stereoisomer in case of n = odd number

**Q3 Calculate all possible optical isomers for following molecules :**

**1.** 

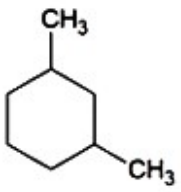

**Sol.** n = 2

Optical isomer = 3

**2.** 

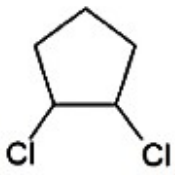

**Sol.** n = 2 chiral centre

Optical isomer = 3

**3.** 


**Sol.** n = 2

Optical isomer = 3

**4.** 


**Sol.** n = 2

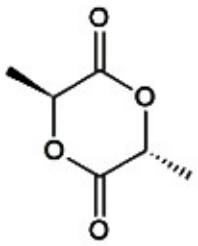
Optical isomer = 3

**Tips and Trick**

Symmetrical molecule → POS / COS

| n | Meso | Optically active | Optically isomer |
|---|---|---|---|
| 2 | 1 | 2 | 3 |
| 3 | 2 | 2 | 4 |
| 4 | 2 | 8 | 10 |
| 5 | 4 | 12 | 16 |

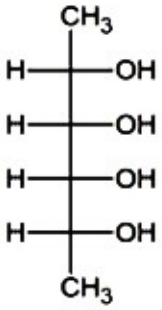
**5.**



**Sol.** n = 2

Optical isomer = 3

**6.**



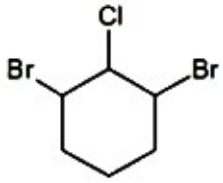

**Sol.** n = 4

$\Rightarrow \quad 2^{4-1} + 2^{\frac{4}{2}-1}$

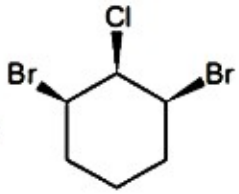
$\Rightarrow \quad 8 + 2 = 10$

**Q4** **Draw all possible optical isomers (consider pseudo chiral centre) :**

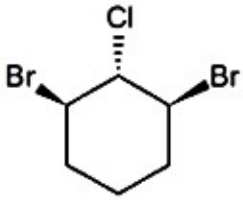
**1.**



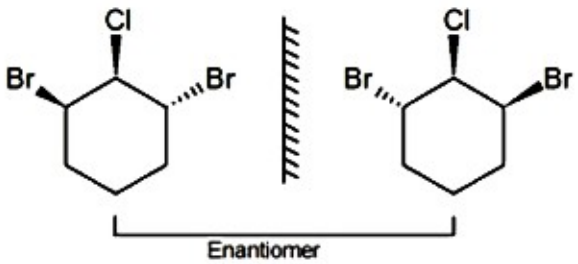

**Sol.**



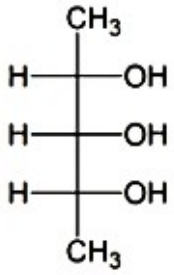

Meso




Meso



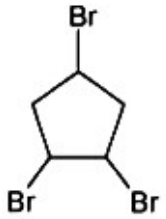
**2.**



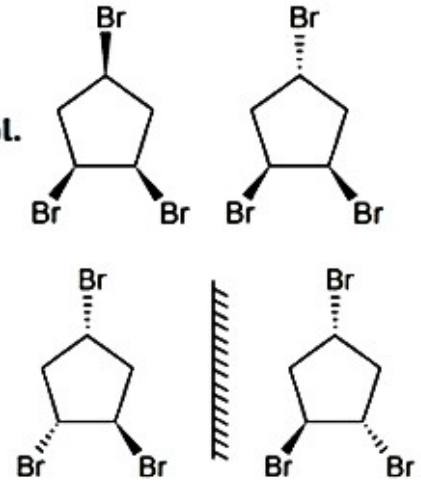

**Sol.** n = 3 (one pseudo)

Optical isomer = 4

**3.**




**Sol.**




Optical isomer = 4

# How to calculate stereo isomer (S.I.)

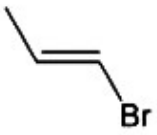
In unsymmetrical,

Stereoisomer = $2^n$

**Q1** **Calculate stereo isomer :**

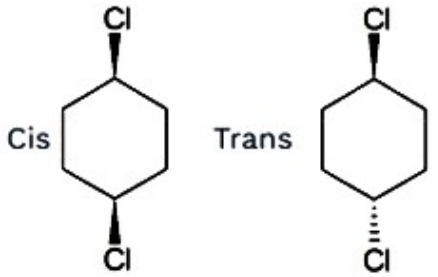
**1.** $CH_3-CH = CH - Br$

**Sol.** $2^1 = 2$

Cis  Trans 

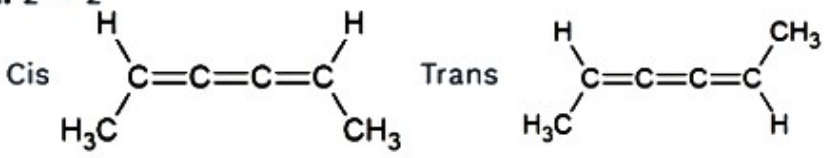
**2.** 

**Sol.** $2^1 = 2$



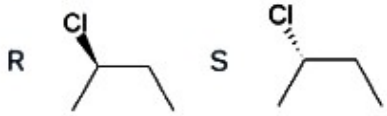

**3.** $CH_3 - CH = C = C = CH - CH_3$

**Sol.** $2^1 = 2$




**4.** $H_3C-CH(Cl)-CH_2-CH_3$

**Sol.** $2^1 = 2$




**Points to remember**

OI → Number of chiral centres

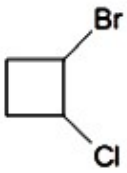
→ Even allene system / spiro / biphenyl

GI → π-bond

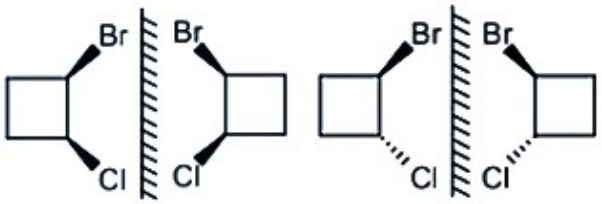
→ Ring

→ Odd system allene / spiro / phenyl

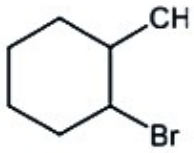
**5.**




**Sol.** $2^2 = 4$

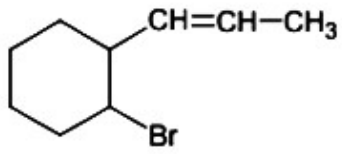



**6.**

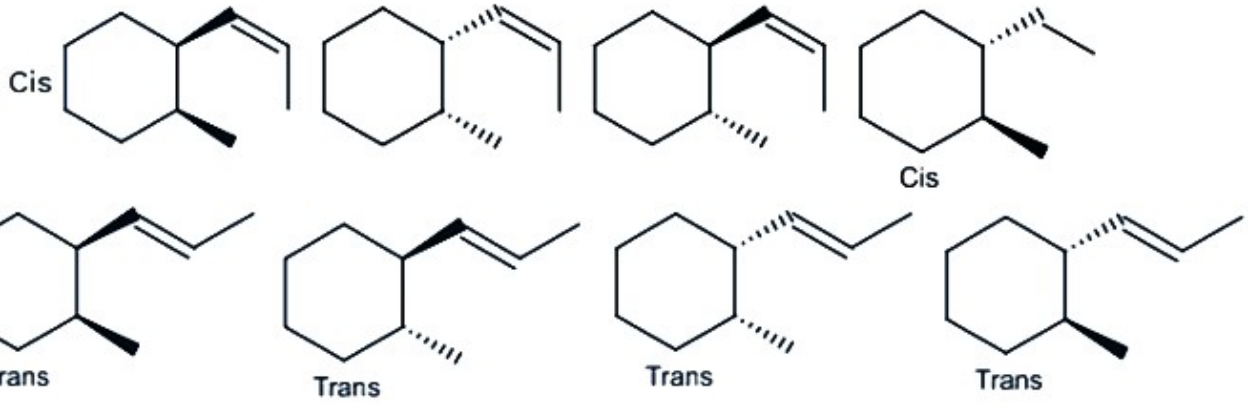



**Sol.** $2^2 = 4$

**7.**

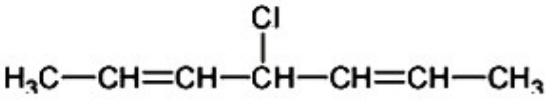



**Sol.** $2^3 = 8$

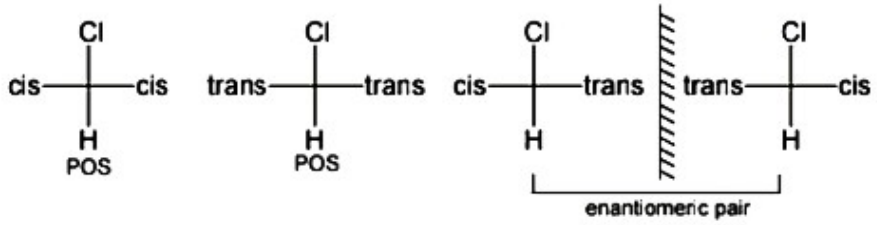


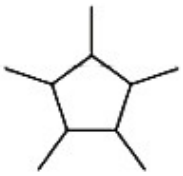

**8.** $H_3C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH-CH=CH_2$

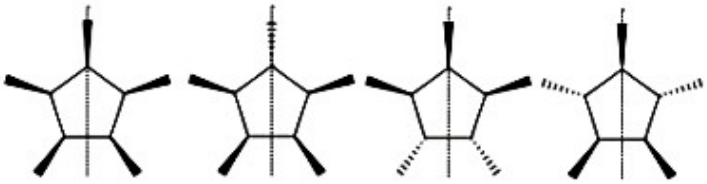
**Sol.** $2^1 = 2$

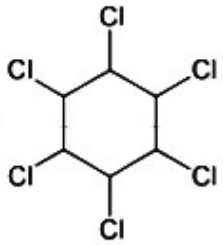
**9.** 

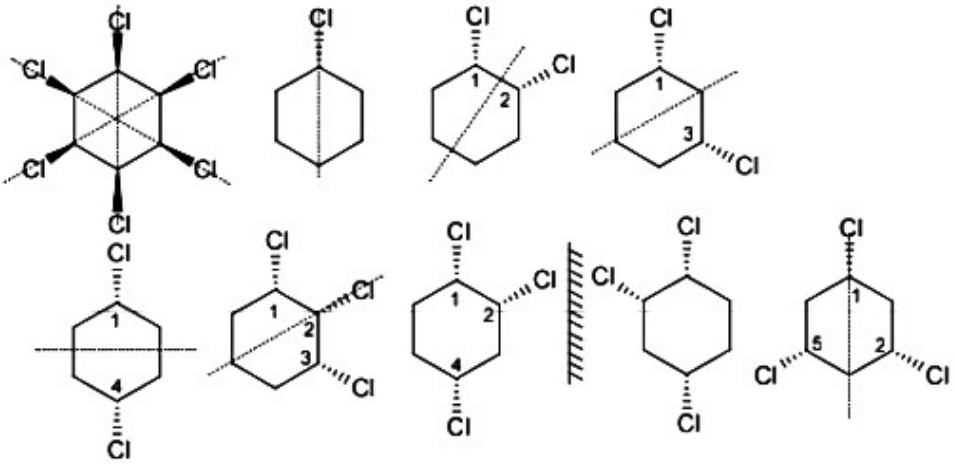
**Sol.** Stereo Isomer = 4




**10.** 

**Sol.** 

**11.** 


**Sol.** 


G.I. = 8

S.I. / O.I. = 9

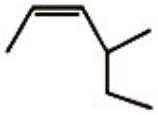
7 meso + 2 enantiomer

**SUBJECTIVE :**

**Q1** **Find :** **Total number of stereoisomers**
**Number of meso compound**
**Number of optically active isomers**

**1.** 

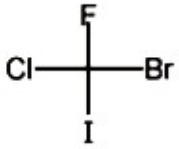
S → 2
M → 0
O.A. → 0

**2.** 

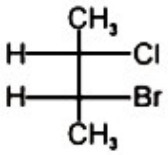
S → 4
M → 0
O.A. → 4

**3.** $CH_3 - CH = C = CH - CH_3$

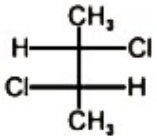
S → 2
M → 0
O.A. → 2

**4.** 


S → 2
M → 0
O.A. → 2

**5.** 


S → 4
M → 0
O.A. → 4

**6.** 


S → 3
M → 1
O.A. → 2

**7.** 2,4-dichloropentane

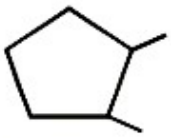
S → 3
M → 1
O.A. → 2

**8.** 2,3-dichloropentane

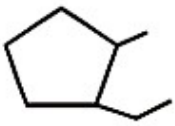
S → 4
M → 0
O.A. → 4

**9.** 1,2-dichloropentane

S → 2
M → 0
O.A. → 2

**10.** 

S → 3
M → 1
O.A. → 2

**11.** 

S → 4
M → 0
O.A. → 4